

Chapter 119

Tracking Nodes

This chapter details the Tracking nodes available in Fusion.

The abbreviations next to each node name can be used in the Select Tool dialog when searching for tools and in scripting references.

For purposes of this document, node trees showing MediaIn nodes in DaVinci Resolve are interchangeable with Loader nodes in Fusion Studio, unless otherwise noted.

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Camera Tracker [CTra]



The Camera Tracker node

Camera tracking is match moving, and a vital link between 2D and 3D, allowing compositors to integrate 3D renders into live-action scenes. The Camera Tracker node is used to calculate the path of a live-action camera and generate a virtual camera in 3D space. This virtual camera's motion is intended to be identical to the motion of the actual camera that shot the scene. Using the calculated position and movement of the virtual camera provides the flexibility to add 3D elements to a live-action scene. Also, the Camera Tracker creates a point cloud in 3D space that can be used to align objects and other 3D models to the live-action scene.

For more information about how to use the Camera Tracker, see *Chapter 86, "3D Camera Tracking,"* in the DaVinci Resolve Reference Manual, or Chapter 26 in the Fusion Reference Manual.



The original video clip



3D objects composited on to the video clip that use the camera tracker to remain aligned with the objects in the frame as the image moves

Inputs

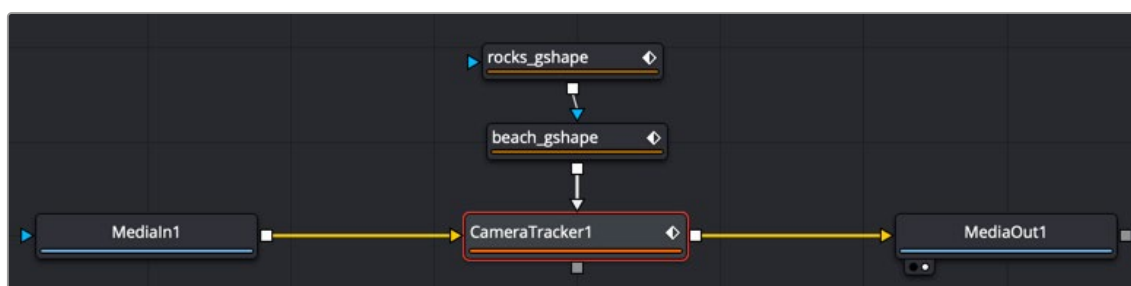
The Camera Tracker has two inputs:

Background: The orange image input accepts a 2D image you want tracked.

Occlusion Mask: The white occlusion mask input is used to mask out regions that do not need to be tracked. Regions where this mask is white will not be tracked. For example, a person moving in front of and occluding bits of the scene may be confusing to the tracker, and a quickly-created rough mask around the person can be used to tell the tracker to ignore the masked-out bits.

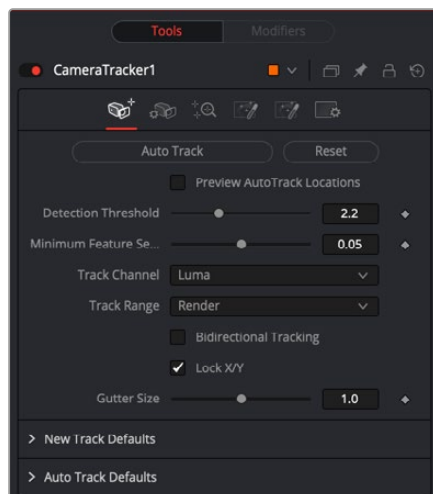
Basic Node Setup

The Camera Tracker background input is used to connect the image you want tracked. Polygon masks can be connected into the occlusion mask input to identify areas the tracker should ignore.



The Camera Tracker with occlusion masks

Inspector



The Camera Tracker tab

Track Tab

The Track tab contains the controls you need to set up an initial analysis of the scene.

Auto Track

Automatically detects trackable features and tracks them through the source footage. Tracks will be automatically terminated when the track error becomes too high, and new tracks are created as needed. The values of the Detection Threshold and Minimum Feature Separation sliders can be used to control the number and distribution of auto tracks.

Reset

Deletes all the data internal to the Camera Tracker node, including the tracking data and the solve data (camera motion path and point cloud). To delete only the solve data, use the Delete button on the Solve tab.

Preview AutoTrack Locations

Turning this checkbox on will show where the auto tracks will be distributed within the shot. This is helpful for determining if the Detection Threshold and Minimum Feature Separation need to be adjusted to get an even spread of trackers.

Detection Threshold

Determines the sensitivity to detect features. Automatically generated tracks will be assigned to the shot, and the Detection Threshold will force them to be either in locations of high contrast or low contrast.

Minimum Feature Separation

Determines the spacing between the automatically generated tracking points. Decreasing this slider causes more auto tracks to be generated. Keep in mind that a large number of tracking points will also result in a lengthier solve.

Track Channel

Used to nominate a color channel to track: red, green, blue, or luminance. When nominating a channel, choose one that has a high level of contrast and detail.

Track Range

Used to determine which frames are tracked:

- **Global:** The global range, which is the full duration of the Timeline.
- **Render:** The render duration set on the Timeline.
- **Valid:** The valid range is the duration of the source media.
- **Custom:** A user-determined range. When this is selected, a separate range slider appears to set the start and end of the track range.

Bidirectional Tracking

Enabling this will force the tracker to track backward after the initial forward tracking. When tracking backward, new tracks are not started but rather existing tracks are extended backward in time. It is recommended to leave this option on, as long tracks help give better solved cameras and point clouds.

Gutter Size

Trackers can become unstable when they get close to the edge of the image and either drift or jitter or completely lose their pattern. The Camera Tracker will automatically terminate any tracks that enter the gutter region. Gutter size is given as a percentage of pattern size. By default, it's 100% of pattern size, so a 0.04 pattern means a 0.04 gutter.

New Track Defaults

There are three methods in which the Camera Tracker node can analyze the scene, and each has its own strengths when dealing with certain types of camera movement.

- **Tracker:** Internally, all the Trackers use the Optical Flow Tracker to follow features over time and then further refine the tracks with the trusted Fusion Tracker or Planar Tracker. The Planar Tracker method allows the pattern to warp over time by various types of transforms to find the best fit. These transforms are:
 - Translation
 - Translation and Rotation Translation, Rotation, and Scale Affine
 - Perspective

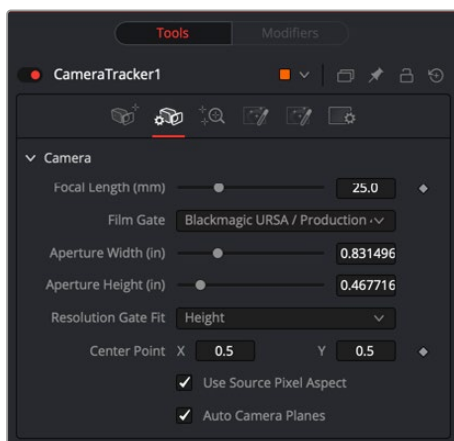
It is recommended to use the default TRS setting when using the Planar Tracker. The Affine and Perspective settings need large patterns in order to track accurately.

- **Close Tracks When Track Error Exceeds:** Tracks will be automatically terminated when the tracking error gets too high. When tracking a feature, a snapshot of the pixels around a feature are taken at the reference time of the track. This is called a pattern, and that same pattern of pixels is searched for at future times. The difference between the current time pattern and the reference time pattern is called the track error. Setting this option higher produces longer but increasingly less accurate tracks.
- **Solve Weight:** By default, each track is weighted evenly in the solve process. Increasing a track's weight means it has a stronger effect on the solved camera path. This is an advanced option that should be rarely changed.

Auto Track Defaults

Set a custom prefix name and/or color for the automatically generated tracks. This custom color will be visible when Track Colors in the Options tab is set to User Assigned.

Camera Tab



The Camera Tracker Camera tab

The controls of the Camera tab let you specify the physical aspects of the live-action camera, which will be used as a starting point when searching for solve parameters that match the real-world camera. The more accurate the information provided in this section, the more accurate the camera solve.

The Camera tab includes controls relating to the lens and gate aspects of the camera being solved for.

Focal Length

Specify the known constant focal length used to shoot the scene or provide a guess if the Refine Focal Length option is activated in the Solve tab.

Film Gate

Choose a film gate preset from the drop-down menu or manually enter the film back size in the Aperture Width and Aperture Height inputs. Note that these values are in inches.

Aperture Width

In the event that the camera used to shoot the scene is not in the preset drop-down menu, manually enter the aperture width (inches).

Aperture Height

In the event that the camera used to shoot the scene is not in the preset drop-down menu, manually enter the aperture height (inches).

Resolution Gate Fit

This defines how the image fits the sensor size. Often, film sensors are sized to cover a number of formats, and only a portion of the sensor area is recorded into an image.

For example, a 16:9 image is saved out of a full aperture-sized sensor.

Typically, fit to Width or Height is the best setting. The other fit modes are Inside, Outside, or Stretched.

Center Point

This is where the camera lens is aligned to the camera. The default is (0.5, 0.5), which is the middle of the sensor.

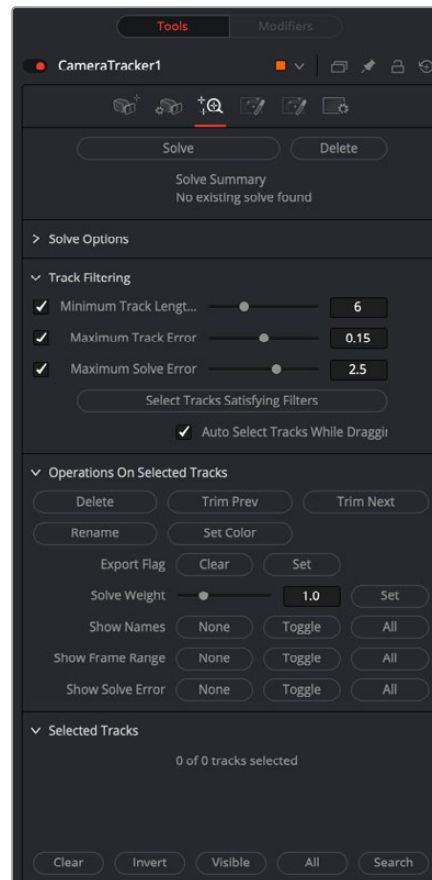
Use Source Pixel Aspect

This will use the squeeze aspect of the pixels that is loaded in the image. HD is square pixels, but NTSC has a pixel aspect ratio of 0.9:1, and Anamorphic CinemaScope is 2:1 aspect. Disabling this option exposes Pixel X and Y number fields where you can customize the source pixel aspect.

Auto Camera Planes

When enabled, the camera's image plane and far plane are automatically moved to enclose the point cloud whenever a solve completes. Sometimes, though, the solver can atypically fling points off really deep into the scene, consequently pushing the image plane very far out. This makes the resulting scene unwieldy to work with in the 3D views. In these cases, disable this option to override this default behavior (or delete the offending tracks).

Solve Tab



The Camera Tracker Solve tab

The Solve tab is where the tracking data is used to reconstruct the camera's motion path along with the point cloud. It is also where clean up of bad or false tracks is done, and other operations on the tracks can be performed, such as defining which marks are exported in the Point Cloud 3D. The markers can also have their weight set to affect the solve calculations.

For example, a good camera solve may have already been generated, but there are not enough locators in the point cloud in an area where an object needs to be placed, so adding more tracks and setting their Solve Weight to zero will not affect the solved camera but will give more points in the point cloud.

Solve

Pressing Solve will launch the solver, which uses the tracking information and the camera specifications to generate a virtual camera path and point cloud, approximating the motion of the physical camera in the live-action footage. The console will automatically open, displaying the progress of the solver.

Delete

Delete will remove any solved information, such as the camera and the point cloud but will keep all the tracking data.

Average Solve Error

Once the camera has been solved, a summary of the solve calculation is displayed at the top of the Inspector. Chief among those details is the Average Solve Error. This number is a good indicator of whether the camera solve was successful. It can be thought of as the difference (measured in pixels) between tracks in the 2D image and the reconstructed 3D locators reprojected back onto the image through the reconstructed camera. Ultimately, in trying to achieve a low solve error, any value less than 1.0 pixels will generally result in a good track. A value between 0.6 and 0.8 pixels is considered excellent.

Clean Tracks by Filter

Clicking this button selects tracks based on the Track Filtering options. If the Auto Delete Tracks By Filter checkbox is activated, the selected tracks will be deleted as well.

Clean Foreground Tracks

Clicking this button makes a selection of the tracks on fast-moving objects that would otherwise cause a high solve error. The selection is determined by the Foreground Threshold slider.

Foreground Threshold

This slider sets the detection threshold for finding the tracks on fast-moving objects. The higher the value, the more forgiving.

Auto Delete Tracks by Filter

With this checkbox enabled, tracks that are selected by the Clean Tracks By Filter button will be deleted. Enable the checkbox, and then press Clean Tracks By Filter. Any track that meets the filtering options is then selected and deleted.

Auto Delete Foreground Tracks

With this checkbox enabled, tracks that are selected by the Clean Foreground Tracks button will be deleted. Enable the checkbox, and then press Clean Foreground Tracks. Any track that meets the foreground threshold criteria is deleted.

Accept Solve Error

This slider sets an acceptable maximum threshold level for the solve error. If the solve error is greater than this value, the Camera Tracker will sweep the focal length setting in an attempt to bring the solve error under the Accept Solve Error value. If the solver cannot find a solution, the Camera Tracker will display a message in the console that the solver failed. If a solution cannot be found, ideally you should try to input the correct focal length or alternatively manually clean some noisy tracks then re-solve.

Auto Select Seed Frames

With this enabled, the Camera Tracker nominates two frames that will be used as a reference for initiating the solve. These two frames are initially solved for and a camera is reconstructed, and then gradually more frames are added in, and the solution is "grown" outward from the seed frames. The choice of seed frames strongly affects the entire solve and can easily cause the solve to fail. Seed frames can be found automatically or defined manually.

Disabling this will allow the user to select their own two frames. Manual choice of seed frames is an option for advanced users. When choosing seed frames, it is important to satisfy two conflicting desires: the seed frames should have lots of tracks in common yet be far apart in perspective (i.e., the baseline distance between the two associated cameras is long).

Refine Focal Length

Enabling this will allow the solver to adjust the focal length of the lens to match the tracking points. You can prevent the focal length being adjusted by setting the Focal Length parameter in the Camera tab.

Enable Lens Parameter

When enabled, lens distortion parameters are exposed to help in correcting lens distortion when solving.

- **Refine Center Point:** Normally disabled, camera lenses are typically centered in the middle of the film gate, but this may differ on some cameras. For example, a cine camera may be set up for Academy 1.85, which has a sound stripe on the left, and shooting super35, the lens is offset to the right.
- **Refine Lens Parameters:** This will refine the lens distortion or curvature of the lens. There tends to be larger distortion on wide angle cameras.

NOTE: When solving for the camera's motion path, a simulated lens is internally created to model lens distortion in the source footage. This simulated lens model is much simpler than real-world lenses but captures the lens distortion characteristics important for getting an accurate camera solve. Two types of distortion are modeled by Camera Tracker:

Radial Distortion: The strength of this type of distortion varies depending on the distance from the center of the lens. Examples of this include pincushion, barrel, and mustache distortion. Larger values correspond to larger lens curvatures. Modeling radial distortion is especially important for wide angle lenses and fisheye lenses (which will have a lot of distortion because they capture 180 degrees of an environment and then optically squeeze it onto a flat rectangular sensor).

Tangential Distortion: This kind of distortion is produced when the camera's imaging sensor and physical lens are not parallel to each other. It tends to produce skew distortions in the footage similar to distortions that can be produced by dragging the corners of a corner pin within Fusion. This kind of distortion occurs in very cheap consumer cameras and is practically non-existent in film cameras, DSLRs, and pretty much any kind of camera used in film or broadcast. It is recommended that it be left disabled.

Enable Lens Parameters

When disabled, the Camera Tracker does not do any lens curvature simulations. This is the default setting and should remain disabled if there is a very low distortion lens or the lens distortion has already been removed from the source clip. Activating the Enable Lens Parameters checkbox determines which lens parameters will be modeled and solved for. Parameters that are not enabled will be left at their default values. The following options are available:

- **Radial Quadratic:** Model only Quadratic radial lens curvature, which is either barrel or pincushion distortion. This is the most common type of distortion. Selecting this option causes the low and high order distortion values to be solved for.
- **Radial Quartic:** Model only Quartic radial lens curvature, which combines barrel and pincushion distortion. This causes the low and high order distortion values to be solved for.
- **Radial & Tangential:** Model and solve for both radial and tangential distortion. Tangential relates to misaligned elements in a lens.
- **Division Quadratic:** Provides a more accurate simulation of Quadratic radial lens curvature. This causes the low and high order distortion values to be solved for.

- **Division Quartic:** Provides a more accurate simulation of Quartic radial lens curvature. This causes the low and high order distortion values to be solved for.

Lower Order Radial Distortion: This slider is available for all simulations. It determines the quadratic lens curvature.

Higher Order Radial Distortion: This slider is available only for Quartic simulations. Determines the quartic lens curvature.

Tangential Distortion X/Y: These sliders are available only for Tangential simulations. Determines skew distortion.

Track Filtering

The Camera Tracker can produce a large number of automatically generated tracks. Rather than spending a lot of time individually examining the quality of each track, it is useful to have some less time-intensive ways to filter out large swaths of potentially bad tracks. The following input sliders are useful for selecting large amounts of tracks based on certain quality metrics, and then a number of different possible operations can be made on them. For example, weaker tracks can be selected and deleted, yielding a stronger set of tracks to solve from. Each filter can be individually enabled or disabled.

Minimum Track Length (Number of Markers)

Selects tracks that have a duration shorter than the slider's value. Short tracks usually don't get a chance to move very far and thus provide less perspective information to the solver than a longer track, yet both short and long tracks are weighted evenly in the solve process, making long tracks more valuable to the solver. Locators corresponding to shorter tracks are also less accurately positioned in 3D space than those corresponding to longer tracks. If the shot has a lot of long tracks, it can be helpful to delete the short tracks. For typical shots, using a value in the range of 5 to 10 is suggested. If there are not a lot of long tracks (e.g., the camera is quickly rotating, causing tracks to start and move off frame quickly), using a value closer to 3 is recommended.

Maximum Track Error

Selects tracks that have an average track error greater than the slider's value. When tracking, tracks are automatically terminated when their track error exceeds some threshold. This auto termination controls the maximum track error, while this slider controls the average track error. For example, tracks following the foliage in a tree tend to be inaccurate and sometimes may be detected by their high average error.

Maximum Solve Error

Selects tracks that have a solve error greater than the slider's value. One of the easiest ways to increase the accuracy of a camera solve is to select the 20% of the tracks with the highest solve error and simply delete them (although this can sometimes make things worse).

Select Tracks Satisfying Filters

Selects the tracks within the scene that meet the above Track Filtering values. Note that when this button is pressed, the tracks that satisfy the filter values are displayed in the Selected Tracks area of the Solve tab and are colored in the viewer. This button is useful when Auto Select Tracks While Dragging Sliders is turned off or if the selection, for example, was accidentally lost by mis-clicking in the viewer.

Auto Select Tracks While Dragging Sliders

When this is ticked on, dragging the above sliders (Minimum Track Length, Maximum Track Error, Maximum Solve Error) will cause the corresponding tracks to be interactively selected in the viewer.

Operations on Selected Tracks

Tracks selected directly in the viewer with the mouse or selected via track filtering can have the following operations applied:

Delete	Removes the tracks from the set. When there are bad tracks, the simplest and easiest option is to simply delete them.
Trim Previous	Cuts the tracked frames from the current frame to the start of the track. Sometimes it can be more useful to trim a track than to delete it. For example, high-quality long tracks that become inaccurate when the feature they are tracking starts to become occluded or when the tracked feature moves too close to the edge of the image.
Trim Next	Cuts the tracked frames from the current frame to the end of the track.
Rename	Replaces the current auto generated name with a new name.
Set Color	Allows for user-assigned color of the tracking points.
Export Flag	This controls whether the locators corresponding to the selected tracks will be exported in the point cloud. By default, all locators are flagged as exportable.
Solve Weight	By default, all the tracks are used and equally weighted when solving for the camera's motion path. The most common use of this option is to set a track's weight to zero, so it does not influence the camera's motion path but still has a reconstructed 3D locator. Setting a track's weight to values other than 1.0 or 0.0 should only be done by advanced users. Onscreen display of track names and values are controlled by these functions:
None	Clears/hides the selected tracks.
Toggle	Swaps the selected tracks and unselected sets.
All	Selects all tracks.
Show Names	Displays the track name. By default, these are numbers.
Show Frame Range	Displays the start and end frame of a track.
Show Solve Error	Displays the amount of solve error of each selected track.

Selected Tracks

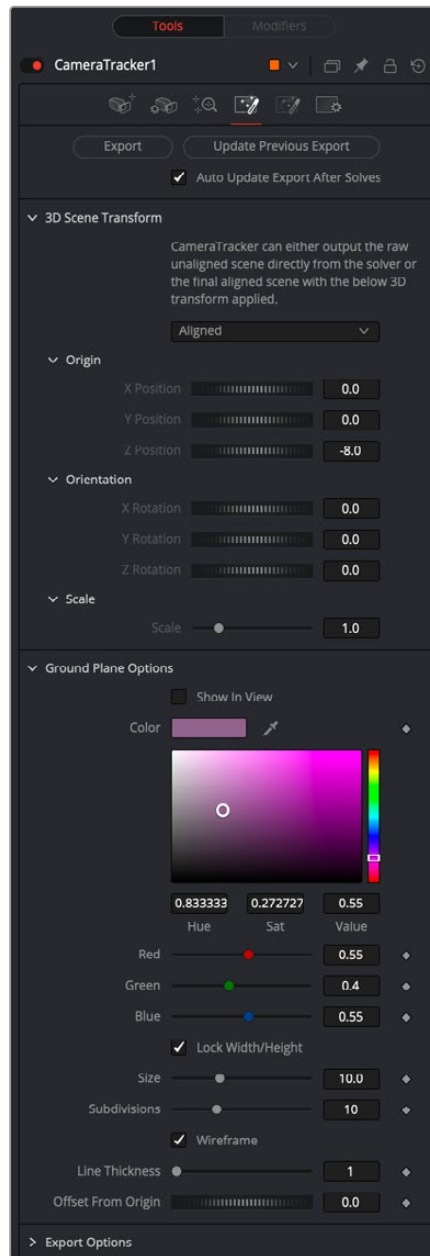
This area displays the properties of a track point or group of points. It has options to:

- **Clear:** Deselects all tracks and clears this area.
- **Invert:** Deselects the current selected tracks and selects the other tracks.
- **Visible:** Selects all the trackers at the current frame.
- **All:** Selects all trackers on all frames.
- **Search:** Selects tracks whose names contain a substring.

TIP: Also select tracks directly in the 2D viewer using the mouse or in the 3D viewer by selecting their corresponding locators in the point cloud.

Export

The Export tab lets you turn the tracked and solved data this node has generated into a form that can be used for compositing.



The Camera Tracker Export tab

Export

The Export button will create a basic setup that can be used for 3D match moving:

- A Camera 3D with animated translation and rotation that matches the motion of the live-action camera and an attached image plane.
- A Point Cloud 3D containing the reconstructed 3D positions of the tracks.
- A Shape 3D set to generate a ground plane.

- A Merge 3D merging together the camera, point cloud, and ground plane. When the Merge 3D is viewed through the camera in a 3D viewer, the 3D locators should follow the tracked footage.
- A Renderer 3D set to match the input footage.

The export of individual nodes can be enabled/disabled in the Export Options tab.

Update Previous Export

When this button is clicked, the previously exported nodes are updated with any new data generated. These previously exported nodes are remembered in the Previous Export section at the bottom of this section. Here's an example of how this is handy:

- 1 Solve the camera and export.
- 2 Construct a complex Node Editor based around the exported nodes for use in set extension.
- 3 The camera is not as accurate as preferred or perhaps the solver is rerun to add additional tracks to generate a denser point cloud. Rather than re-exporting the Camera 3D and Point Cloud 3D nodes and connecting them back in, just press the Update Previous Export button to “overwrite” the existing nodes in place.

Automatically Update Previous Export After Solves

This will cause the already exported nodes (Camera 3D, Point Cloud 3D, Lens Distort, Renderer 3D, and the ground plane) to auto update on each solve.

3D Scene Transform

Although the camera is solved, it has no idea where the ground plane or center of the scene is located. By default, the solver will always place the camera in Fusion's 3D virtual environment so that on the first frame it is located at the origin (0, 0, 0) and is looking down the Z-axis. You have the choice to export this raw scene without giving the Camera Tracker any more information, or you can set the ground plane and origin to simplify your job when you begin working in the 3D scene. The 3D Scene Transform controls provide a mechanism to correctly orient the physical ground plane in the footage with the virtual ground plane in the 3D viewer. Adjusting the 3D Scene Transform does not modify the camera solve but simply repositions the 3D scene to best represent the position of the live-action camera.

NOTE: If you export the scene and then make changes in the 3D Scene Transform, it is important to manually click Update Previous Export to see the results in the exported nodes.

Aligned/Unaligned

The Aligned/Unaligned menu locks or unlocks the origin and ground plane settings. When set to Unaligned, you can select the ground plane and origin either manually or by selecting locators in the viewer. When in unaligned mode, a 3D Transform control in the 3D viewer can be manually manipulated to adjust the origin.

Once alignment of the ground plane and origin has been completed, the section is locked by switching the menu to Aligned.

Set from Selection

When set to unaligned, buttons labeled Set from Selection are displayed under the Origin, Orientation, and Scale sections. Clicking these buttons takes the selecting locators in the viewer and aligns the ground plane or origin based on the selection.

For instance, to set the ground plane, do the following:

- 1 After solving, set the 3D Scene Transform menu to Unaligned.
- 2 Find a frame where the ground plane is at its largest and clearest point.
- 3 In the viewer, drag a selection rectangle around all the ground plane locators.
- 4 Hold Shift and drag again to add to the selection.
- 5 In the Orientation section, make sure the Selection Is menu correctly matches the orientation of the selected locators.
- 6 Click the Set from Selection button located under the Orientation parameters.
- 7 Set the 3D Scene Transform menu back to Aligned.

To get the best result when setting the ground plane, try to select as many points as possible belonging to the ground and having a wide separation.

TIP: When selecting points for the ground plane, it is helpful to have the Camera Tracker node viewed in side-by-side 2D and 3D views. It may be easier to select tracks belonging to the ground by selecting tracks from multiple frames in the 2D viewer rather than trying to box select locators in the 3D viewer.

Setting the origin can help you place 3D objects in the scene with more precision. To set the origin, you can follow similar steps, but only one locator is required for the origin to be set. When selecting a locator for the origin, select one that has a very low solve error.

Ground Plane Options

These controls let you adjust the ground plane for the scene, which is a crucial step in making sure the composite looks correct.

Color	Sets the color of the ground plane.
Size	Controls how big the ground plane can be set.
Subdivision Level	Shows how many polygons are in the ground plane.
Wireframe	Sets whether the ground plane is set as wireframe or solid surface when displayed in 3D.
Line Thickness	Adjusts how wide the lines will draw in the viewer.
Offset	By default, the center of the ground plane is placed at the origin (0, 0, 0). This can be used to shift the ground plane up and down along the Y-axis.

Export Options

Provides a checkbox list of what will be exported as nodes when the Export button is pressed. These options are: Camera, Point Cloud, Ground Plane, Renderer, Lens Distortion, and Enable Image Plane in the camera.

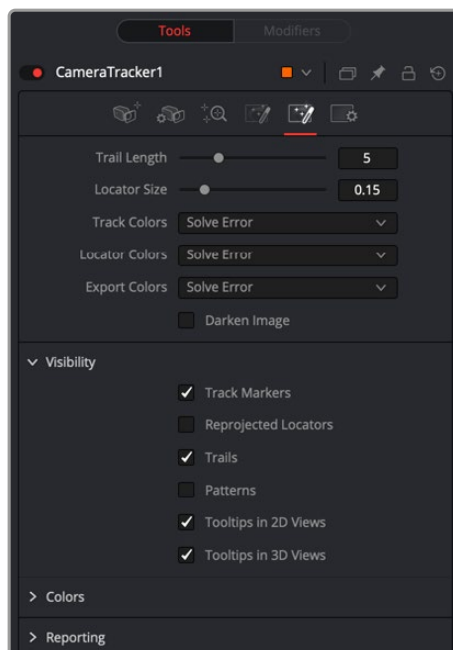
The Animation menu allows you to choose between animating the camera and animating the point cloud. Animating the camera leaves the point cloud in a locked position while the camera is keyframed to match the live-action shot. Animating the point cloud does the opposite. The camera is locked in position while the entire point cloud is keyframed to match the live-action shot.

Previous Export

When the Update Previous Export button is clicked, the previously exported nodes listed here are updated with any new data generated (this includes the camera path and attributes, the point cloud, and the renderer).

Options

The Options tab lets you customize the Camera Tracker's onscreen controls so you can work most effectively with the scene material you have.



The Camera Tracker Options tab

Trail Length

Displays trail lines of the tracks overlaid on the viewer. The amount of frames forward and back from the current frame is set by length.

Location Size

In the 3D viewer, the point cloud locators can be sized by this control.

Track Colors, Locator Colors, and Export Colors each have options for setting their color to one of the following: User Assigned, Solve Error, Take From Image, and White.

Track Colors	The onscreen tracks in the 2D viewer.
Locator Colors	The point cloud locators in the 3D viewer.
Export Colors	The colors of the locators that get exported within the Point Cloud node.

Darken Image

Dims the brightness of the image in viewers to better see the overlaid tracks. This affects both the 2D and 3D viewers.

Visibility

Toggles which overlays will be displayed in the 2D and 3D viewers. The options are: Tracker Markers, Trails, Tooltips in the 2D viewer, Tooltips in the 3D viewer, Reprojected Locators, and Tracker Patterns.

Colors

Sets the color of the overlays.

- **Selection Color:** Controls the color of selected tracks/locators.
- **Preview New Tracks Color:** Controls the color of the points displayed in the viewer when the Preview AutoTrack Locations option is enabled.
- **Solve Error Gradient:** By default, tracks and locators are colored by a green-yellow-red gradient to indicate their solve error. This gradient is completely user adjustable.

Reporting

Outputs various parameters and information to the Console.

Understanding Camera Tracking

On large productions, camera tracking or 3D match moving is often handed over to experts who have experience with the process of tracking and solving difficult shots. There is rarely a shot where you can press a couple of buttons and have it work perfectly. It does take an understanding of the whole process and what is essential to get a good solved track.

The Camera Tracker must solve for hundreds of thousands of unknown variables, which is a complex task. For the process to work, it is essential to get good tracking data that exists in the shot for a long time. False or bad tracks will skew the result. This section explains how to clean up false tracks and other techniques to get a good solve.

Workflow

Getting a good solve is a repeated process.

Track > Solve > Refine Filters > Solve > Cleanup tracks > Solve > Cleanup from point cloud > Solve > Repeat.

Initially, there are numerous tracks, and not all are good, so a process of filtering and cleaning up unwanted tracks to get to the best set is required. At the end of each cleanup stage, pressing Solve ideally gives you a progressively lower solve error. This needs to be below 1.0 for it to be good for use with full HD content, and even lower for higher resolutions. Refining the tracks often but not always results in a better solve.

False Tracks

False tracks are caused by a number of conditions, such as moving objects in a shot, or reflections and highlights from a car. There are other types of false tracks like parallax errors where two objects are at different depths, and the intersection gets tracked. These moiré effects can cause the track to creep. Recognizing these False tracks and eliminating them is the most important step in the solve process.

Track Lengths

Getting a good set of long tracks is essential; the longer the tracks are, the better the solve. The Bi-Directional tracking option in the Tracker tab is used to extend the beginning of tracks in time. The longer in time a track exists and the more tracks that overlap in time of a shot, the more consistent and accurate the solve.

Seed Frames

Two seed frames are used in the solve process. The algorithm chooses two frames that are as far apart in time yet share the same tracks. That is why longer tracks make a more significant difference in the selection of seed frames.

The two Seed frames are used as the reference frames, which should be from different angles of the same scene. The solve process will use these as a master starting point to fit the rest of the tracks in the sequence.

There is an option in the Solve tab to Auto Detect Seed Frames, which is the default setting and most often a good idea. However, auto detecting seed frames can make for a longer solve. When refining the Trackers and re-solving, disable the checkbox and use the Seed 1 and Seed 2 sliders to enter the previous solve's seed frames. These seed frames can be found in the Solve Summary at the top of the Inspector after the initial solve.

Refine Filters

After the first solve, all the Trackers will have extra data generated. These are solve errors and tracking errors.

Use the refine filters to reduce unwanted tracks, like setting minimum tracker length to eight frames. As the value for each filter is adjusted, the Solve dialog will indicate how many tracks are affected by the filter. Then Solve again.

Onscreen Culling

Under the Options tab, set the track to 20. This will display each track on footage with +-20 frame trail. When scrubbing/playing through the footage, false tracks can be seen and selected onscreen, and deleted by pressing the Delete key. This process takes an experienced eye to spot tracks that go bad. Then Solve again.

You can view the exported scene in a 3D perspective viewer. The point cloud will be visible. Move and pan around the point cloud, select and delete points that seem to have no inline with the image and the scene space. Then Solve again.

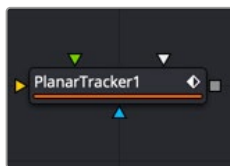
Repeat the process until the solve error is below 1.0 before exporting.

Common Controls

Settings Tab

The Settings tab in the Inspector is also duplicated in other Tracking nodes. These common controls are described in detail in the following "The Common Controls" section.

Planar Tracker Node [PTRA]



The Planar Tracker node

The Planar Tracker node is designed to solve a match-moving problem that commonly comes up during post-production. As an example, live-action footage can often contain a planar surface such as a license plate or a road sign that needs new numbers in the license plate or a new city's name on the road sign. Often, the problem is that the camera is moving in the shot, so the license plate or road sign is continuously changing perspective. You cannot just merge a new license plate over the existing one without accounting for the perspective distortions. A time-intensive way to solve this problem would be to use a Corner Pin node and manually keyframe the four corners. The Planar Tracker automates this keyframing process and tracks the perspective distortions of a planar surface over time. That tracking data is then applied with those same perspective distortions to a different foreground.

For more information on using the Planar Tracker, see *Chapter 83, "Planar Tracking,"* in the DaVinci Resolve Reference Manual, or Chapter 23 in the Fusion Reference Manual.

TIP: Part of using a Planar Tracker is also knowing when to give up and fall back to using Fusion's Tracker node or to manual keyframing. Some shots are simply not trackable, or the resulting track suffers from too much jitter or drift. The Planar Tracker is a useful time-saving node in the artist's toolbox, but while it may track most shots, it is not a 100% solution.

What the Planar Tracker Saves

While the Planar Tracker does save the resulting final track in the composition on disk, it does not save temporary tracking information, such as the individual point trackers (compared with the Camera Tracker, which does save the individual point trackers). Some consequences of this include:

- The point trackers no longer appear in the viewer when a comp containing a Planar Tracker node is saved and reloaded.
- Tracking may not be resumed after a comp containing a Planar Tracker node has been saved and reloaded. In particular, this also applies to auto saves. For this reason, it is good to complete all planar tracking within one session.
- The size of composition files is kept reasonable (in some situations, a Planar Tracker can produce hundreds of megabytes of temporary tracking data).
- Saving and loading of compositions is faster and more interactive.

Inputs

The Planar Tracker has four inputs:

Background: The orange background image input accepts a 2D image with the planar surface to be tracked.

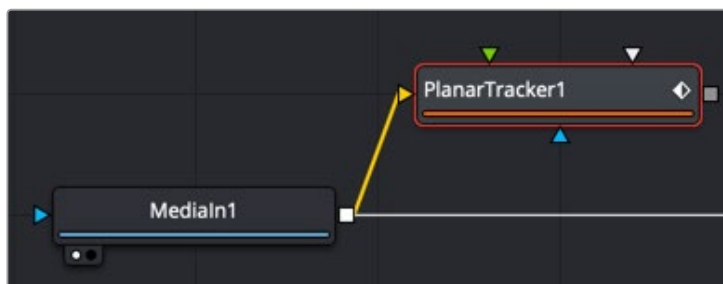
Corner Pin 1: The green corner pin 1 input accepts a 2D image to be pinned on top of the background. There may be multiple corner pin inputs, named Corner Pin 1, Corner Pin 2, etc.

Occlusion Mask: The white occlusion mask input is used to mask out regions that do not need to be tracked. Regions where this mask is white will not be tracked. For example, a person moving in front of and occluding bits of the pattern may be confusing the tracker, and a quickly-created rough rotomask around the person can be used to tell the tracker to ignore the masked-out bits.

Effect Mask: The blue input is for a mask shape created by polylines, basic primitive shapes, paint strokes, or bitmaps from other tools. Connecting a mask to this input limits the output of the Planar Tracker to certain areas.

Basic Node Setup

A basic Planar Tracker setup consists of just two nodes: a MediaIn connects to the background input and the Planar Tracker can be used as a separate branch from the rest of the node tree. Once the tracking is completed, a Planar Transform node should be generated to use the planar tracking data.



A Planar Tracker can be isolated on its own branch of a node tree.

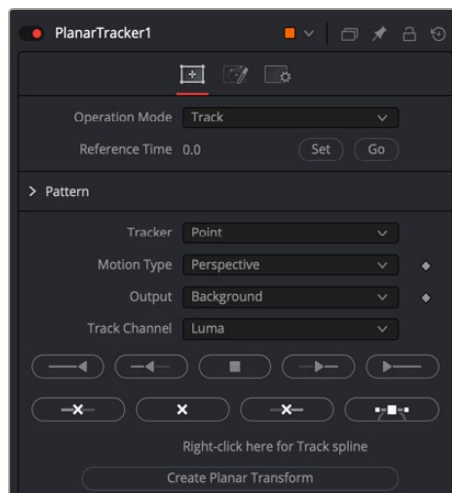
A Typical Planar Tracker Workflow

The following steps outline the workflow with the Planar Tracker:

- 1 Remove lens distortion: The more lens distortion in the footage, the more the resulting track will slide and wobble.
- 2 Connect footage: Connect a Loader or MediaIn node that contains a planar surface to the orange background input and view the Planar Tracker node in a viewer.
- 3 Select a reference frame: Move to a frame where the planar surface to be tracked is not occluded and click the Set button to set this as a reference frame.
- 4 Choose the pattern: In the viewer, make sure the onscreen controls are visible, and draw a polygon around the planar surface you want to track. This is called the “pattern.” In most cases, this will probably be a rectangle, but an arbitrary closed polygon can be used. The pixels enclosed by this region will serve as the pattern that will be searched for on other frames. Note that it is important that the pattern is drawn on the reference frame. Do not confuse the pattern with the region to corner pin (which always has four corners and is separately specified in Corner Pin mode).
- 5 Adjust render range: In the Keyframes Editor, adjust the render range to match the range of frames where the planar surface is visible.

- 6 Adjust track options: Frequently changed options include Tracker, Motion Type, and Track Channel.
- 7 Mask out occluders: If moving objects partially cover up the planar surface, you may wish to connect an occlusion mask to the Planar Tracker. When using the Hybrid tracker, providing a mask to deal with occluding objects is strongly recommended, while with the Point tracker it is recommended to try tracking without a mask.
- 8 Track: Click the Go button to return to the reference frame. Press the Track To End button and wait for the track to complete. Click the Go button to return to the reference frame again. Press the Track To Start button and wait for the track to complete. Note that the tracks in the viewer are not selectable or deletable as they are in a Camera Tracker.
- 9 Check track quality: Visually inspect the track to see how accurate it is. Does it stick to the surface? Switching to Steady mode can help here.
- 10 Use the track: At this point, in most cases, you will create a Planar Transform node from the Inspector and use it to apply the tracked perspective distortion onto masked images. If the image you are applying the tracking data to is a full frame unmasked clip, you can use the Steady, Corner Pin, and Stabilize operation modes in the Planar Tracker.

Inspector



The Planar Tracker Controls tab

Controls tab

The Controls tab contains controls for determining how the Planar Tracker will be used, setting the reference frame, and initiating the track.

Operation Mode

The Operation Mode menu selects the purpose of the Planar Tracker node. The Planar Tracker has four modes of operation:

- **Track:** Used to isolate a planar surface and track its movement over time. Then, you can create a Planar Transform node that uses this data to match move another clip in various ways.
- **Steady:** After analyzing a planar surface, this mode removes all motion and distortions from the planar surface, usually in preparation for some kind of paint or roto task, prior to “unsteadying” the clip to restore the motion.

- **Corner Pin:** After analyzing a planar surface, this mode computes and applies a matching perspective distortion to a foreground image you connect to the foreground input of the Planar Tracker node and merges it on top of the tracked footage.
- **Stabilize:** After analyzing a planar surface, this mode allows smoothing of a clip's translation, rotation, and scale over time. This is good for getting unwanted vibrations out of a clip while retaining the overall camera motion that was intended.

The last three modes (Steady, Corner Pin, and Stabilize) use the tracking data produced in Track mode.

NOTE: None of the operations can be combined together. For example, both Corner Pin and Stabilize cannot be done at the same time, nor can a track be done while in corner pinning mode.

Reference Time

The Reference Time determines the frame where the pattern is outlined. It is also the time from which tracking begins. The reference frame cannot be changed once it has been set without destroying all pre-existing tracking information, so scrub through the footage to be tracked and choose carefully. The reference frame must be chosen carefully to give the best possible quality track.

You choose a reference frame by moving the playhead to an appropriate frame and then clicking the Set button to choose that frame.

Pattern Polygon

You specify which region of the image you want to track by drawing a polygon on the reference frame. Typically, when you first add a Planar Tracker node, you are immediately ready to start drawing a polygon in the viewer, so it's best to do this right away. When choosing where to draw a polygon, make sure the region selected belongs to a physically planar surface in the shot. In a pinch, a region that is only approximately planar can be used, but the less planar the surface, the poorer the quality of the resulting track.

As a rule of thumb, the more pixels in the pattern, the better the quality of the track. In particular, this means the reference frame pattern should be:

- As large as possible.
- As much in frame as possible.
- As unoccluded as possible by any moving foreground objects.
- At its maximal size (e.g., when tracking an approaching road sign, it is good to pick a later frame where it is 400 x 200 pixels big rather than 80 x 40 pixels).
- Relatively undistorted (e.g., when the camera orbits around a flat stop sign, it is better to pick a frame where the sign is face-on parallel to the camera rather than a frame where it is at a highly oblique angle).

If the pattern contains too few pixels or not enough trackable features, this can cause problems with the resulting track, such as jitter, wobble, and slippage. Sometimes dropping down to a simpler motion type can help in this situation.

After you've drawn a pattern, a set of Pattern parameters lets you transform and invert the resulting polygon, if necessary.

Track Mode

Track mode is unlike the other three options in the Operation menu in that it is the only option that initiates the planar tracking. The other modes use the tracking data generated by the Track mode.

Tracker

There are two available trackers to pick from:

- **Point:** Tracks points from frame to frame. Internally, this tracker does not actually track points-per-se but rather small patterns like Fusion's Tracker node. The point tracker possesses the ability to automatically create its internal occlusion mask to detect and reject outlier tracks that do not belong to the dominant motion. Tracks are colored green or red in the viewer, depending on whether the point tracker thinks they belong to the dominant motion or they have been rejected. The user can optionally supply an external occlusion mask to further guide the Point tracker.
- **Hybrid Point/Area:** Uses an Area tracker to track all the pixels in the pattern. Unlike the Point tracker, the Area tracker does not possess the ability to automatically reject parts of the pattern that do not belong to the dominant motion, so you must manually provide it with an occlusion mask. Note that for performance reasons, the Hybrid tracker internally first runs the Point tracker, which is why the point tracks can still be seen in the viewer.

There is no best tracker. They each have their advantages and disadvantages:

Artist Effort (occlusion masks): The Point tracker will automatically create its internal occlusion mask. However, with the Hybrid tracker, you need to spend more time manually creating occlusion masks.

Accuracy: The Hybrid tracker is more accurate and less prone to wobble, jitter, and drift since it tracks all the pixels in the pattern rather than a few salient feature points.

Speed: The Hybrid tracker is slower than the Point tracker.

In general, it is recommended to first quickly track the shot with the Point tracker and examine the results. If the results are not good enough, then try the Hybrid tracker.

Motion Type

Determines how the Planar Tracker internally models the distortion of the planar surface being tracked. The five distortion models are:

- Translation.
- Translation, Rotation (rigid motions).
- Translation, Rotation, Scale (takes squares to squares, scale is uniform in x and y).
- Affine - includes translation, rotation, scale, skew (maps squares to parallelograms).
- Perspective (maps squares to generic quadrilaterals).

Each successive model is more general and includes all previous models as a special case.

When in doubt, choose Perspective for the initial track attempt. If the footage being tracked has perspective distortions in it, and the Planar Tracker is forced to work with a simpler motion type, this can end up causing the track to slide and wobble.

Sometimes with troublesome shots, it can help to drop down to a simpler motion model—for example, when many track points are clustered on one side of the tracked region or when tracking a small region where there are not many trackable pixels.

Output

Controls what is output from the Planar Tracker node while in the Track operation mode.

- **Background:** Outputs the input image unchanged.
- **Background - Preprocessed:** The Planar Tracker does various types of preprocessing on the input image (e.g., converting it to luma) before tracking. It can be useful to see this when deciding which track channel to choose.
- **Mask:** Outputs the pattern as a black and white mask.
- **Mask Over Background:** Outputs the pattern mask merged over the background.

Track Channel

Determines which image channel in the background image is tracked. It is good to pick a channel with high contrast, lots of trackable features, and low noise. Allowed values are red, green, blue, and luminance.

Tracking Controls

These controls are used to control the Tracker. Note that while tracking, only track to a new frame if the current frame is already tracked or it is the reference frame.

- **Track to start:** Tracks from the current frame backward in time to the start (as determined by the current render range).
- **Step tracker to previous frame:** Tracks from the current frame to the previous frame.
- **Stop tracking:** Stops any ongoing tracking operations.
- **Step tracker to next frame:** Tracks from the current frame to the next frame.
- **Track to end:** Tracks from the current frame forward in time to the end (as determined by the current render range).
- **Trim to start:** Removes all tracking data before the current frame.
- **Delete:** Deletes all tracking data at all times. Use this to destroy all current results and start tracking from scratch.
- **Trim to end:** Removes all tracking data after the current frame. This can be useful, for example, to trim the end of a track that has become inaccurate when the pattern starts to move off frame.

Show Splines

This button to the right of the “Trim to end” button opens the Spline Editor and shows the splines associated with the Planar Tracker node. This can be useful for manually deleting points from the Track and Stable Track splines.

Right-Click Here for Track Spline

While tracking, a spline containing 4 x 4 matrices at each keypoint is created. This is known as the “Track spline” or just “Track” for short. These matrices completely describe the distortions of the tracked pattern.

Create Planar Transform

After tracking footage, this button can be pressed to create a Planar Transform node on the Node Editor. The current tracking data is embedded in the Planar Transform node so that it can replicate the planar distortions tracked by the Planar Tracker node. Unless you are compositing a full frame foreground that matches the same dimensions as the raster, it is best to create a Planar Transform and use it to apply motion to the foreground.

Steady Mode

In Steady mode, the Planar Tracker transforms the background plate to keep the pattern as motionless as possible. Any leftover motion is because the Planar Tracker failed to follow the pattern accurately or because the pattern did not belong to a physically planar surface.

Steady mode is not very useful for actual stabilization, but is useful for checking the quality of a track. If the track is good, during playback the pattern should not move at all while the rest of the background plate distorts around it. It can be helpful to zoom in on parts of the pattern and place the mouse cursor over a feature and see how far that feature drifts away from the mouse cursor over time.

Steady Time

This is the time where the pattern's position is snapshotted and frozen in place. It is most common to set this to the reference frame.

Invert Steady Transform

This causes the Planar Tracker node to reverse the effects of the steady transform. This means two Planar Tracker nodes connected back to back with the second set to invert should give back the original image. If you place an effects node in between the two, then the effect will be locked in place. This should only be used to accomplish effects that cannot be done through corner pinning, since it involves two resamplings, causing softening of the background image.

Clipping Mode

Determines what happens to the parts of the background image that get moved off frame by the steady transform:

- **Domain:** The off frame parts are kept.
- **Frame:** The off frames parts are thrown away.

Domain mode is useful when Steady mode is being used to “lock” an effect to the pattern. As an example, consider painting on the license plate of a moving car. One way to do this is to use a Planar Tracker node to steady the license plate, then a Paint node to paint on the license plate, and then a second Planar Tracker to undo the steady transform. If the Clipping mode is set to Domain, the off frame parts generated by the first Planar Tracker are preserved so that the second Planar Tracker can, in turn, map them back into the frame.

Corner Pin Mode

In Corner Pin mode, one or more textures can be attached to a previously tracked planar surface and undergo the same perspective distortions as the surface.

The corner pin workflow with Planar Tracker is:

- 1 **Track:** select a planar surface in the shot that you wish to attach a texture to or replace the texture on. Track the shot (see the tracking workflow in the Track section).
- 2 **Switch the Operation Mode to Corner Pin:** When Corner Pin mode is entered from Track mode, the pattern polygon is hidden and a corner pin control is shown in the viewer.
- 3 **Connect in the texture:** In the Node Editor, connect the output of the MediaIn node containing the texture to the Corner Pin 1 input on the Planar Tracker node.
- 4 **Adjust corner pin:** Drag the corners of the corner pin in the viewer until the texture is positioned correctly. Sometimes the Show Grid option is useful when positioning the texture. Additionally, if it helps to position it more accurately, scrub to other times and make adjustments to the corner pin.
- 5 **Review:** Play back the footage and make sure the texture “sticks” to the planar surface.

Merge Mode

Controls how the foreground (the corner pinned texture) is merged over the background (the tracked footage). If there are multiple corner pins, this option is shared by all of them. There are four options to pick from:

- BG only
- FG only
- FG over BG
- BG over FG

Number of Corner Pins

Use the + and - buttons to increase or decrease the number of corner pins. Each time an additional corner pin is created, a corresponding input appears on the node in the Node Editor.

Corner Pin 1 Input Group

Each corner pin has a group of related inputs:

- **Enable:** Controls the visibility of the corner pin in the viewer.
- **Show Grid:** Shows a grid over the corner pin. This can be useful when positioning the corners.
- **Merge Options:** Controls merging of corner pin texture over the background. (See the documentation for the Merge node.)
- **Reference Time Positions:** The positions of the four corners at the reference time. If the track was not perfect, these positions can be animated to make adjustments on top of the track.

Stabilize Mode

Stabilize mode is used to smooth out shakiness in the camera by applying a transform that partially counteracts the camera shake. This stabilizing transform (contained in the Stable Track spline) is computed by comparing neighboring frames.

NOTE: Stabilize mode only smooths out motions, while Steady mode tries to completely “lock off” all motion.

Be aware that the Planar Tracker stabilizes based on the motion of the pattern, so it is essential to choose the pattern carefully. If the motion of the pattern does not represent the motion of the camera, then there may be unexpected results. For example, if tracking the side of a moving truck and the camera is moving alongside it, the Planar Tracker smooths the combined motion of both the truck and the mounted camera. In some cases, this is not the desired effect. It may be better to choose the pattern to be on some fixed object like the road or the side of a building, which would result in smoothing only the motion of the camera.

One unavoidable side effect of the stabilization process is that transparent edges appear along the edges of the image. These edges appear because the stabilizer does not have any information about what lies off frame, so it cannot fill in the missing bits. The Planar Tracker node offers the option to either crop or zoom away these edges. When filming, if the need for post-production stabilization is anticipated, it can sometimes be useful to film at a higher resolution (or lower zoom).

The Planar Tracker Stabilization Workflow

- 1 **Track:** Select a roughly planar region that represents the motion that you want to stabilize. Track the shot (see the tracking workflow in the Track section).
- 2 **Switch the Operation Mode to Stabilize:** Until a stabilization is computed, the Planar Tracker will just output the input footage.
- 3 **Adjust stabilization options:** Frequently changed options are Parameters to Smooth and Smoothing Radius.
- 4 **Compute stabilization:** Press the Compute Stabilization button and wait for the stabilization computations to finish. Play back the output of the Planar Tracker node to see the effects of the stabilization. Notice that transparent edges have been introduced around the edges of the image by the stabilization transform.
- 5 **Refine:** Adjust the stabilization options and recompute the stabilization as many times as desired.
- 6 **Handle transparent edges (optional):** Set the Frame Mode to either Zoom or Crop as desired and then click the Auto Zoom or Auto Crop button. Play back the footage to observe the effects. If there is too much zoom or the image has been cropped too small, try reducing the amount of smoothing.

Parameters to Smooth

Specify which of the following parameters to smooth:

- X Translation
- Y Translation
- Rotation
- Scale

Smoothing Window

When stabilizing a particular frame, this determines how the contributions of neighboring frames are weighted. Available choices are Box and Gaussian.

Smoothing Radius (Frames)

Determines the number of frames whose transforms are averaged together to compute the stabilization. A larger Smoothing Radius results in more smoothing but introduces more transparent edges.

Compute Stabilization

Clicking this button runs the stabilizer, overwriting the results of any previous stabilization. As soon as the stabilization is finished, the output of the Planar Tracker node will be immediately updated with the stabilization applied.

NOTE: The stabilizer uses the Track spline (created by the tracker) to produce the Stable Track spline. Both of these splines' keyframes contain 4 x 4 matrices, and the keyframes are editable in the Spline Editor.

Clipping Mode

Determines what happens to the parts of the background image that get moved off frame by the stabilization:

- **Domain:** The off frame parts are kept.
- **Frame:** The off frames parts are thrown away.

Frame Mode

This controls how transparent edges are handled. The available options include:

- **Full:** Do nothing. Leaves the transparent edges as is.
- **Crop:** Crops away the transparent edges. When this option is selected, the size of Planar Tracker's output image is smaller than the input image. No image resamplings occur. In Crop mode, use the Auto Crop button or manually adjust the crop window by changing the X Offset, Y Offset, and Scale sliders.

Auto Crop Button: When this button is clicked, the Planar Tracker will examine all the frames and pick the largest possible crop window that removes all the transparent edges. The computed crop window will always be centered in frame and pixel aligned. When clicked, Auto Crop updates the X/Y Offset and Scale sliders.

- **Zoom:** Scales the image bigger until the transparent edges are off frame. Choosing this option causes an image resampling to occur. The downside of this approach is that it reduces the quality (slightly softens) of the output image. In Zoom mode, use the Auto Zoom button or manually adjust the zoom window by changing the X Offset, Y Offset, and Scale sliders.

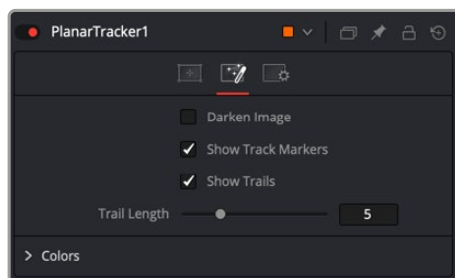
Auto Zoom: When this button is clicked, the Planar Tracker will examine all the frames and pick the smallest possible zoom factor that removes all the transparent edges. The computed zoom window will always be centered in frame. When clicked, Auto Zoom updates the X/Y Offset and Scale sliders.

Right-Click Here for Stable Track Spline

Right-clicking over this label provides access to a spline whose keyframes contain 4 x 4 matrices which in turn represent the stabilization transforms. This is mostly here for completeness and for advanced users.

Options Tab

These controls affect the look of onscreen controls in the viewer.



The Planar Tracker Options tab

Darken Image

Darkens the image while in Track mode in order to better see the controls and tracks in the viewer. The Shift+D keyboard shortcut toggles this.

Show Track Markers

Toggles the display of the dots marking the location of trackers at the current time.

Show Trails

Toggles the display of the trails following the location of trackers.

Trail Length

Allows changing the length of tracker trails. If the pattern is moving very slowly, increasing the length can sometimes make the trails easier to follow in the viewer. If the pattern is moving very fast, the tracks can look like spaghetti in the viewer. Decreasing the length can help.

Inlier/Outlier Colors

When tracking, the tracker analyzes the frame and detects which of the multitudinous tracks belong to the dominant motion and which ones represent anomalous, unexplainable motion. By default, tracks belonging to the dominant motion are colored green (and are called inliers) and those that do not belong are colored red (and are called outliers). Only the inlier tracks are used when computing the final resulting track.

Common Controls

Settings Tab

The Settings tab in the Inspector is also duplicated in other Tracking nodes. These common controls are described in detail at the end of this chapter in “The Common Controls” section.

Surface Tracker [SFt]



The Surface Tracker node

Surface Tracker Node Introduction

The Surface Tracker is used for applying textures and effects to flexible or deformable moving surfaces, like cloth and skin. The Surface Tracker ensures that the folding and deforming of the texture or effect is realistically matched to the folding and deforming of the surface that it is attached to in the image.

The Surface tracker is a complex effect, with many different options and controls to create the most realistic composites possible. But in all cases you simply move from left to right, using the buttons in the effect's settings.

- 1 Click the Bounds button, and click on the screen to define the area on the surface where you wish to apply the new texture. The bounds should select one surface at a time and not go over/on object edges.
- 2 Click on the Mesh button to define the control points along the natural folds and curves of the surface within the bounds.
- 3 Click on the Track button to analyze the motion of the surface over time, using the mesh you just set.
- 4 Click on the Result button to composite the texture on top of the moving mesh.

Across all the tabs in the Surface Tracker, the Tracked Range shows the start and end frame numbers of your clip that contain tracking data. Any existing tracking data will be reset if you modify the bounds, or mesh layout, and you will need to re-track it.



The Surface Tracker in action, overlaying the logo (left), and warping and conforming it realistically to the folds of the shirt (right).

For more information on using the Surface Tracker, see *Chapter 168, "Resolve FX Transform."*

Inputs

There are four inputs to the Surface Tracker node.

Input: The yellow input accepts the media with the surface that you want to track and composite onto.

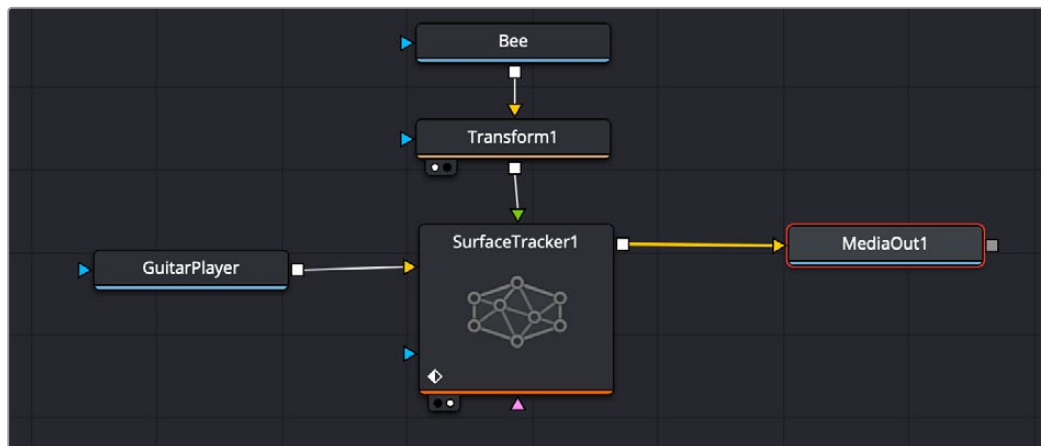
Overlay: The green input accepts still graphics, video, or the output of a node tree. This is the image that you want to overlay onto the surface.

Occlusion Mask: The pink occlusion mask input is used to mask out regions that do not need to be tracked. Regions where this mask is white will not be tracked. For example, a person moving in front of and occluding bits of the scene may be confusing to the tracker, and a quickly-created rough mask around the person can be used to tell the tracker to ignore the masked-out bits.

Effect Mask: The optional blue effect mask input accepts a mask shape created by polylines, basic primitive shapes, paint strokes, or bitmaps from other tools. Connecting a mask to this input limits the filter to only those pixels within the mask. An effects mask is applied to the tool after the tool is processed.

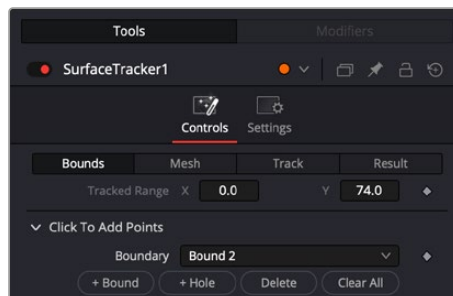
Basic Node Setup

Connect the media with the surface you want to composite onto to the main yellow input. Connect the media with the graphic or video that you want to overlay onto the surface to the green overlay input. Optionally, connect effect or occlusion masks to their respective inputs, if needed.



A simple node tree showing video clip of a guitar player, using the surface tracker to overlay a small graphic of a bee on his shirt.

Inspector



The Bounds tab in the Surface Tracker controls

Bounds

This is where you are prompted to draw an area on the image that you want to apply the texture to. Ideally, you want to choose a frame in the clip that has the surface you want to track as close and as flat-on to the camera as possible. This frame becomes the default reference frame for the effect. Simply click on the image to make multiple points around the area you want to define (a minimum of three points are required), and the effect will connect them automatically as a polygon. Boundary points should be on the surface for tracking, not around it. If points are accidentally placed on object edges or the background instead, it will cause errors in the tracking.

You may add multiple boundaries and multiple holes to the same image, and all boundaries will join together into one shape before subtracting the holes. The purpose of multiple boundaries and holes are to allow you to mask out complex surfaces with regions not for tracking. For example, you can

make a boundary around someone's face, but use multiple holes to exclude the eyes and mouth as those change appearance quickly as the subject blinks and talks, making them exceptionally difficult to track.

Multiple boundaries are designed to work on the same surface; using multiple boundaries on different surfaces within the image will cause poor results. Separate surfaces are best tracked by using multiple instances of the Surface Tracker instead.

The boundary is not useful for rotoscoping objects, only for analyzing the motion of a surface. Rotoscoping of foreground objects can be done as described in the Tracking Using a Mask section below.

- **Click to Add Points:** These are the controls used to make boundary areas on the image.

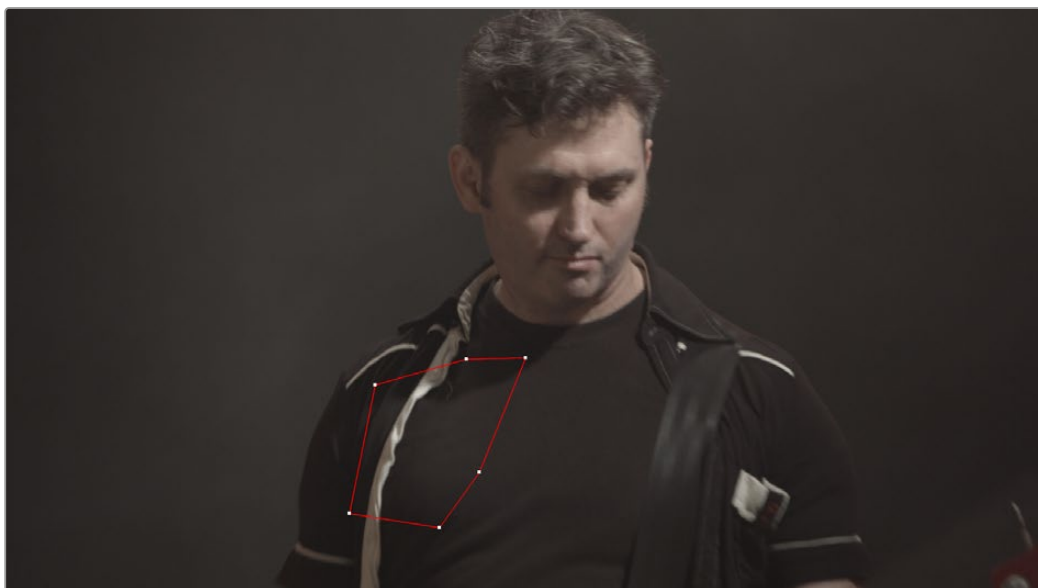
Boundary: The currently selected boundary area on the image. You can change the current boundary or hole by using the drop-down menu. If there are issues with the boundary (i.e., crossed edges, too few points, etc.), an error message will appear next to the boundary name.

+ **Bound:** Adds a boundary polygon to the image. You can have multiple boundaries defined for the same surface, and they will be joined together into one shape.

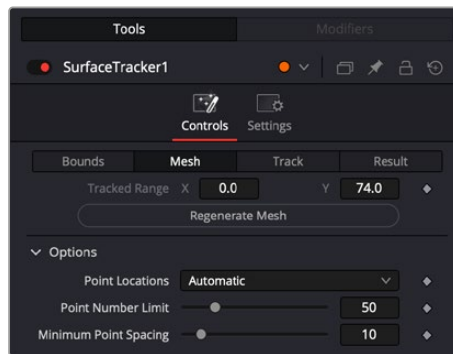
+ **Hole:** Adds an area to exclude tracking from the surface. Used to remove overly complex and changing parts of a surface from the effect's calculations.

Delete: Lets you delete the boundary or hole selected in the Boundary field. Currently, this operation can not be undone, so be careful around this button.

Clear All: Lets you delete all boundaries and holes in the effect.



A simple boundary drawn on his shirt in the viewer



The Mesh tab in the Surface Tracker controls

Mesh

With the boundaries now defined, you can establish the mesh over the surface to be tracked. The initial mesh serves as the starting point for tracking, so for best results, the mesh should be placed when your surface is as flat-on and closest to the camera as possible.

The mesh can be manually adjusted by dragging current points (multiple points can be moved by holding command and dragging an area around the points) or clicking on an area to place new ones. Option-clicking a point will remove it from the mesh. Manually adding mesh points can be useful if there are flexible portions of the surface known to be a point where the texture should fold but which is not initially identified when placing the mesh. For example, an extended arm may not have a mesh point active at the elbow, but with a mouse click (and a basic knowledge of human anatomy), you can manually add one there. Another use case would be to add points to an area with little detail, but you know will develop detail as it rotates into view. Placing mesh points in blank areas that show little detail will not improve results and may actually degrade them, so very few points on smooth areas is desirable. A good rule of thumb is that if more than five points are created inside the boundaries, the mesh may be good enough for tracking. Points cannot be added or deleted after tracking.

A good mesh is a sign of good tracking as they go hand in hand. If your mesh has few points detected and the results don't track tightly, you can artificially improve your results. For example, make sure that the content is clearly visible. Tracking and point detection can fail if the surface is too bright (HDR), too dark, or low contrast due to working in log. To improve results, you can temporarily add a brightness/contrast node before the Surface Tracker to adjust contrast manually, then run the Surface Tracker. Simply disable the brightness/contrast node again after the tracking has finished.

- **Regenerate Mesh:** This button recreates the mesh at the current frame and settings and discards any manual changes that have been made.

Point Locations: These controls determine which mode is used to generate the mesh points.

Automatic: The Surface Tracker automatically locates the best details for tracking, and places mesh points on those details. In most cases, Automatic will give you the best results.

Point Number Limit: Lets you set the maximum amount of mesh points to place. The number of mesh points affects how the overlay will flex and fold, and should be set by the expected flexibility of the surface. They do not directly affect tracking speed.

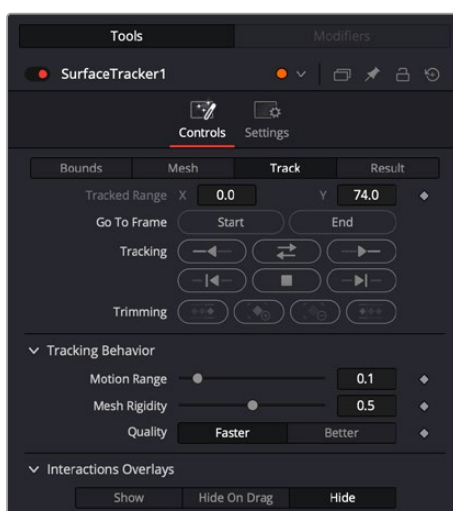
Minimum Point Spacing: Lets you set the minimum distance between mesh points. This prevents mesh points from bunching together too tightly in highly detailed areas and interfering with each other.

Uniform: Creates a regular grid of points in the boundary and relies on internal shape models to deform the grid during tracking. This may give better results than Automatic mode, if your surface is highly detailed.

- **Horizontal Spacing:** Adjusts the horizontal distance between adjacent points in the uniform mesh.
- **Vertical Spacing:** Adjusts the vertical distance between adjacent points in the uniform mesh.



The automatic mesh applied to the boundary. The bee is now composited onto the shirt, but it does not look very realistic yet.



The Track tab in the Surface Tracker controls

Track

This section of the effect tracks the motion of the mesh points over time. Use the Tracking Direction controls to complete the tracking procedure across the clip. After the tracking is complete, the effect does no more analysis, and any additional contrast assistance from previous nodes can be discarded after this step. For simple cases, the tracking may run completely without user intervention, but for more complex cases, a number of additional controls are used to refine the track.

Fixing a broken track will require some manual adjustments to both the mesh and the tracking data. At any time while tracking, mesh points can be adjusted either one by one, or with the lasso selection tool (Command - Drag) around the entire mesh or just parts of it. The lasso selection tool also has the ability to stretch, scale, and rotate the mesh points, making it easier to manually deform and place the mesh exactly where you want it to be, if for some reason the automatic tracking fails. However, additional mesh points cannot be added or deleted at this stage, without first resetting the tracking.

Manually editing the mesh points at a frame only affects that frame directly; it does not affect neighboring tracking data. However, it does affect neighboring interpolated positions, which makes it good for tracking from that point onwards. For example:

- Tracking is good until frame 150 when a few points separate from the rest, disturbed by a lighting change. At frame 150 they can be manually moved back into place, then if tracking is resumed they will continue with their corrected positions.
- Tracking exists until frame 150, after which there is no data for a few frames until it resumes at 160. Editing the points at either frames 150 or 160 will cause the intermediate frames to be interpolated with these changes applied.
- Tracking exists up to frame 150 and at frame 160 onwards, frames 151-159 are interpolated. At frame 155 some points are adjusted. This creates tracking data at this frame, and now frames 151-154 interpolate from the last tracked frame to this current one, and then frames 156-159 from the current to the next tracked frame.

Go to Start: This button will snap the playhead to the first frame that has been tracked, or the frame where the boundary was created.

Go To End: This button will snap the playhead to the last frame that has been tracked.

Tracking Direction Controls: These controls perform the tracking analysis in the direction indicated. They are from left to right: Track Backwards, Track Backwards One Frame, Pause Tracking, Track Forwards Then Reverse, Track Forwards One Frame, Track Forwards. Using these controls, you can set the track across the whole clip or limit the tracking to just a section of it.

Trimming: These controls allow you to manipulate the track data after it's been run.

- **Remove Tracking Data Before This Frame:** Clears all the tracking data before the current frame.
- **Track From Last Known Match:** This button will match the tracking data to the current frame and connect it to the last frame where tracking data exists. If there has been too much motion between those times, you can move the points approximately into place and try matching again. For example, the tracking occurs without issue until frame 150, but something moving across the scene causes an error in the track for six frames (151-157). The tracking data can be paused before it hits these frames, or these frame's tracking data can be deleted after the fact. Then at frame 158 you can press this button to continue the track from the last known good track at frame 150.
- **Remove Tracking Data At This Frame:** Clears the tracking data at the current frame.
- **Remove Tracking Data After This Frame:** Clears all the tracking data after the current frame.

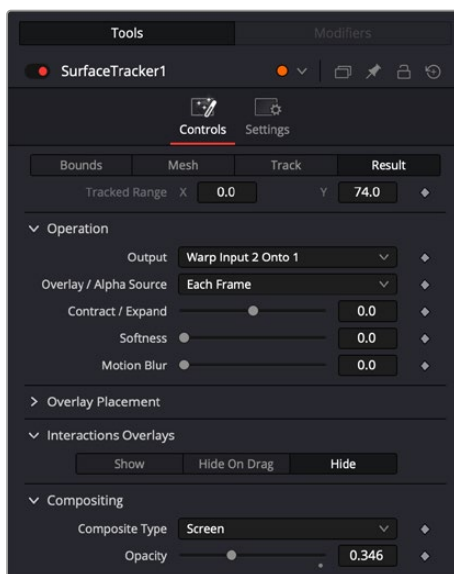
Tracking Behavior: These options allow you adjust the way the tracking operates.

- **Motion Range:** Controls how far to search for matches. Increase this slider if the surface is moving too quickly and the tracking fails to find it.
- **Mesh Rigidity:** Controls how flexible the mesh is. Reduce this slider to make the mesh points more flexible. This gives you best results if the clip is noise-free and warping smoothly. Increase this slider to get a "stiffer" tracked surface with less internal warping.
- **Quality:** The tracker can be very computationally intensive; it is recommended to use the Faster setting while making adjustments to the mesh, then use Better for the final track once all the parameters have been set correctly.

Interaction Overlays: These buttons control the visibility of the mesh overlays on the Viewer.

- **Show:** Shows the mesh at all times.
- **Hide on Drag:** Shows the mesh until a point is dragged, then it is hidden, allowing you to see the surface underneath for fine adjustments.
- **Hide:** Hides the mesh at all times.

NOTE: The Surface Tracker creates data at each frame of the source clip. If the source clip is not at the same frame rate as the timeline, tracking may seem to stutter or skip frames. You can “bake” retiming into a clip by right-clicking on it on the Edit page and choosing “Render In Place” to create a result with exactly one frame for every timeline frame.



The Result tab in the Surface Tracker controls

Result

Once the tracking data is complete, the Result section lets you determine what and how to composite a video layer on top of the moving surface.

Output: There are six options for what to do with the resulting tracking data.

- **Warp Input 2 Onto 1:** Warps the Surface Tracker’s overlay input to the shape and position of the tracked mesh and composites it onto the clip.
- **Warp Input 2 Onto Blank:** Warps the Surface Tracker’s overlay input to the shape and position of the tracked mesh and passes the Alpha channel information, allowing you to perform the actual composite in another node or effect.
- **Warp Input 1’s Alpha:** Applies the results of the Surface Tracker to the Effect Mask input.
- **Create Mesh Alpha Mask:** Produces an Alpha mask, which follows the position and shape of the outline of the mesh. All inputs and alphas are ignored.
- **Stabilize-Warp Input 1:** Performs an inverse warp to the Reference Frame (again, this should be a frame that ideally is as close to and flat-on to the camera as possible). The result is a stationary image within the boundary with all the lighting changes and details of the surface’s motion shown on it. This now stabilized image can be modified with any of DaVinci Resolve’s existing tools, bypassing any issues caused by this area’s motion in the original video clip. Once modified, you warp the result back onto the original footage by using a copy of the original Surface Tracker node but with the Rewarp Stabilized Clip setting here instead.

- **Rewarp Stabilized Clip:** Composites the output of the Stabilize-Warp Input 1 selection back onto the original video clip, after it's been modified. This option is designed to be used in conjunction with a copy of a Surface node using Stabilize-Warp Input 1. It seamlessly blends whatever additional modifications that were made between the two Surface Trackers back together.

Overlay/Alpha Source: Allows you to select which frames of the overlay or alpha are used to create the result.

- **First Frame:** Useful for freezing content from an overlay clip to use a still frame as a static texture.
- **Reference Frame:** Starts at the currently defined reference frame (see below), instead of at frame 0. The reference frame is the frame which the mesh overlay is initially positioned in its “un-warped” state, and the positioning controls are active.
- **Each Frame:** Treats the overlay or alpha as a clip, warping each frame in turn. This is necessary for animated overlays or re-warping stabilized content.

Contract/Expand: Allows you to add or remove pixels from the edges of the overlay or alpha. It is important to note that the expanded area is not tracked and will not be stabilized to the content.

Softness: Blurs the edge of the Expand/Contract tool to blend more realistically into the scene.

Motion Blur: This slider lets you add motion blur to the overlay to preserve the illusion that it was actually captured with the scene.

Overlay Placement: These controls let you set the initial transform options for the image you want to overlay on the mesh.

- **Go To Reference:** Moves the playhead to the selected reference frame.
- **Set to Current:** Sets the current frame under the playhead as the reference frame.
- **Reference Frame:** Shows the frame number at which the overlay is originally positioned and un-warped (the point from which the warp to every other frame is calculated). The positioning controls appear on screen when the clip is at the reference frame. A frame where the surface's view is largest and least warped is the best starting point for rendering onto other frames.
- **Positioning:** Lets you choose how to adjust the overlay position.

Sliders: Choosing this option exposes a set of transform sliders (X/Y position, zoom, rotate, etc.) that let you manipulate the overlay's initial position.

Interactive Canvases: Choosing this option exposes draggable frame controls in the Viewer, allowing you to transform the overlay using mouse controls. The Viewers shows a large grid with 9 regions; you can drag in any of them to move, skew, or warp the overlay.

Interactive Pins: Adjusting the image in this mode is done by manually placing control points, called pins, in the Timeline Viewer. Adding one pin only gives you position control. At least two points are required for scaling and rotation. Dragging on one of the pins scales or rotates the image around the other pin. Using three pins, you can create perspective distortions by dragging any one of the pins. You can add up to four pins for unique corner pinning distortions.

- **Reset Position:** Resets the on-screen overlay back to its default state.

Interaction Overlays: These buttons control the visibility of the mesh overlays on the Viewer.

- **Show:** Shows the mesh at all times.
- **Hide on Drag:** Shows the mesh until a point is dragged, then it is hidden, allowing you to see the surface underneath for fine adjustments.
- **Hide:** Hides the mesh at all times.

Compositing: Lets you set the composite blend mode for the overlay operation. For more detailed information on each blend mode, see Chapter 50, “Compositing and Transforms in the Timeline.”

- **Composite Type:** A drop-down menu of all the possible composite operations.
- **Opacity:** This slider controls the transparency of the overlay.



The completed surface tracking composites the bee graphic onto his shirt. The composite type was screen, and the opacity reduced to blend into the shirt. The bee moves realistically as the shirt folds and deforms as he plays.

Tracker [TRA]



The Tracker node

Tracker Node Introduction

The Tracker is used to detect and follow one or more pixel patterns across frames in moving video. The tracking data can then be used to control the position or values of other nodes in the composition (for example, the center of a Light Rays node). Additionally, trackers can be used to stabilize an image or to apply destabilization to one image based on the motion of another.

For more information, see *Chapter 82, "Using the Tracker Node,"* in the DaVinci Resolve Reference Manual or Chapter 22 in the Fusion Reference Manual.

Inputs

The Tracker has three inputs:

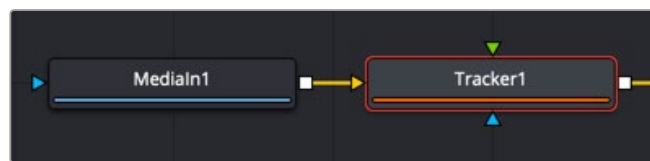
Background: The orange image input accepts the main 2D image to be tracked.

Foreground: The optional green foreground accepts a 2D image to be merged on top of the background as a corner pin or match move.

Effect Mask: The blue input is for a mask shape created by polylines, basic primitive shapes, paint strokes, or bitmaps from other tools. Connecting a mask to this input limits the tracking to certain areas.

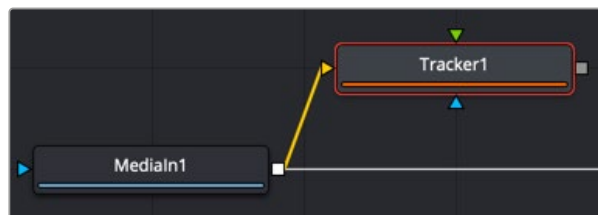
Basic Node Setup

Tracker nodes can be applied inline with other nodes or as a branch from the clip you want to track. When used inline, an image can be stabilized by connecting it to the orange background input. After the image is tracked, setting the Tracker's Operation menu to Match Move will apply a stabilization to the connected image.



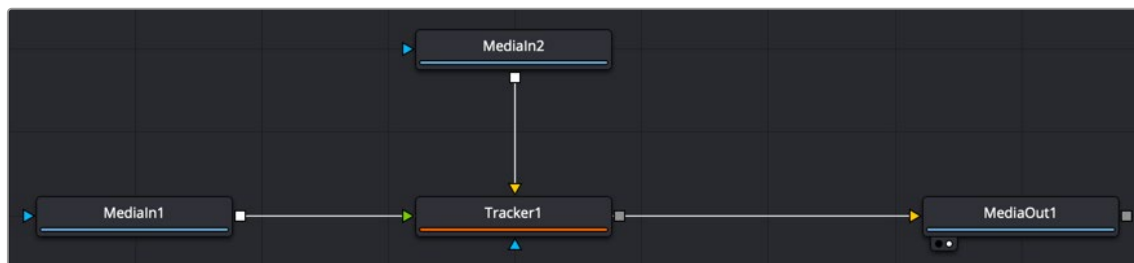
A Tracker node connected inline to an image for stabilization

When used as an offshoot from the node tree, an image can be tracked and then that tracking data is published for use on another node somewhere else in the node tree. The output of the Tracker does not need to connect to another node. The tracking data is published and can be used via the Connect To contextual menu.



A Tracker node branched from the node tree

The Tracker can also work as a replacement for a Merge tool in match-moving setups. Below, the Tracker tracks the image connected to the orange background input and applies the tracking data to the image connected to the foreground input. The same foreground-over-background merge capabilities are available in the Tracker node.



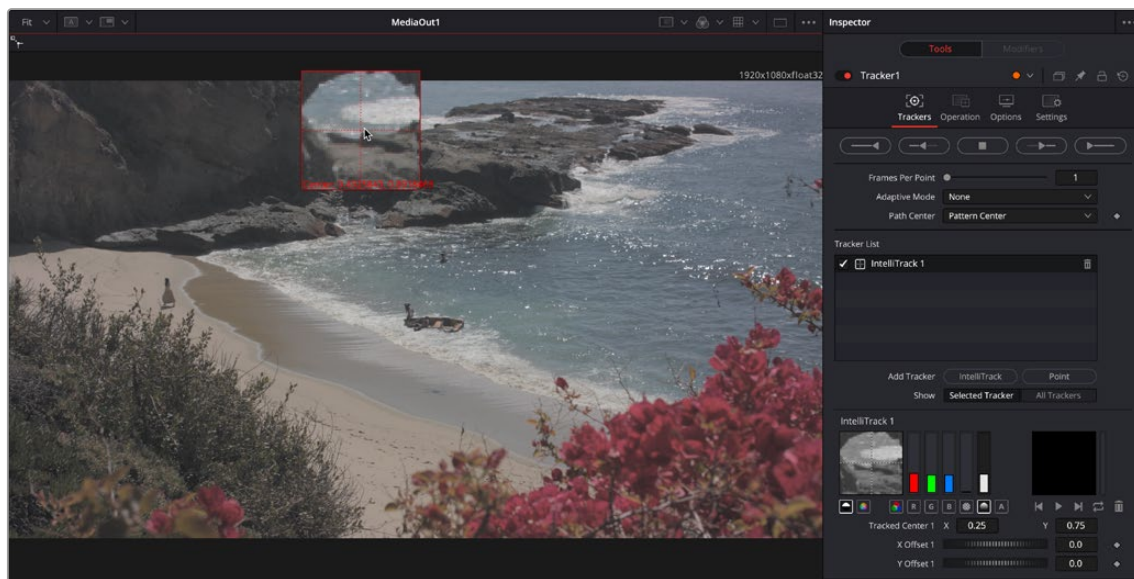
A Tracker node set up to apply a match move to the foreground input

IntelliTrack AI Point Tracker

The Fusion Tracker node now defaults to using IntelliTrack, the new DaVinci Neural Engine tracker. Using AI makes tracking easier and more precise, especially when tracking many points.

IntelliTrack, like other AI tools, doesn't use predetermined rulesets or heuristics. Instead, it is trained on real world examples. This makes it less likely to fail in scenarios like tracking a subject behind brief occlusions. For most cases it will be more precise and more robust than the normal point tracker.

All the IntelliTrack analysis happens behind the scenes, and there is no additional user input required, other than to choose the IntelliTrack setting, set the tracking crosshairs over high contrast areas on the same moving subject, and then press one of the tracker direction controls.



Adding a Tracking Point in the IntelliTrack mode in the Fusion Tracker

NOTE: The original Point tracker is still available to use by choosing the Point button.

Legacy Point Tracker Onscreen Controls

Though Fusion now defaults to IntelliTrack, if you wish to continue to use the old Point tracker, it is still available by clicking the Point button. Each pattern in the tracker has its own set of onscreen controls used to select the pixels in the image to be tracked. These controls are visible in the viewers whenever you select a tracker in the node tree.



Tracker onscreen controls

Pattern Rectangle

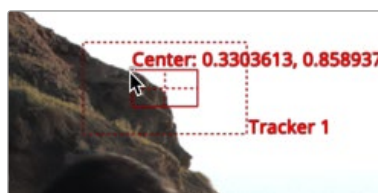
In the viewer, the tracker displays a solid-line red rectangle called the pattern rectangle. Every pixel within the rectangle makes up the pattern used for tracking. You can resize the pattern if necessary by dragging on the rectangle's borders.



A pattern rectangle identifies the area to track

Search Rectangle

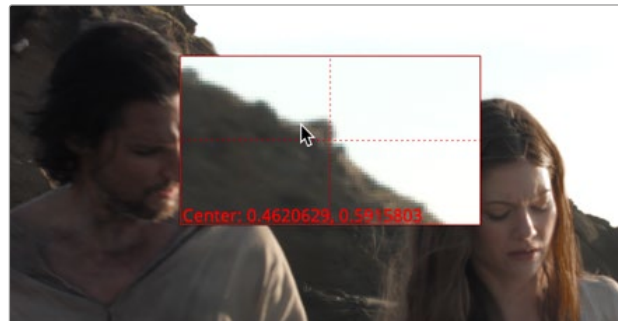
Whenever the mouse moves over the pattern rectangle, a second rectangle with a dashed outline appears. The dashed outline represents the search area, which determines how far away from the current pattern the Tracker looks in the next frame. The search area should always be larger than the pattern, and it should be large enough to encompass the largest frame-to-frame movement in the scene. Faster moving objects require larger search areas, and slower moving objects can get away with smaller search areas. The larger the search area, the longer it takes to track, so try not to make the search area larger than necessary. If the selected Tracker has a custom name, the name of that Tracker is displayed as a label at the bottom right of the search area rectangle.



The search rectangle is the area searched from frame to frame to locate the pattern.

Repositioning the Tracker

The pattern rectangle has a small handle in the upper-left corner. Dragging on the handle repositions the pattern. An enlarged view of the pattern is displayed under your mouse pointer to assist with the precise positioning of the pattern. This thumbnail disappears when the mouse button is released. You can adjust the magnification ratio in the Inspector's Options tab.

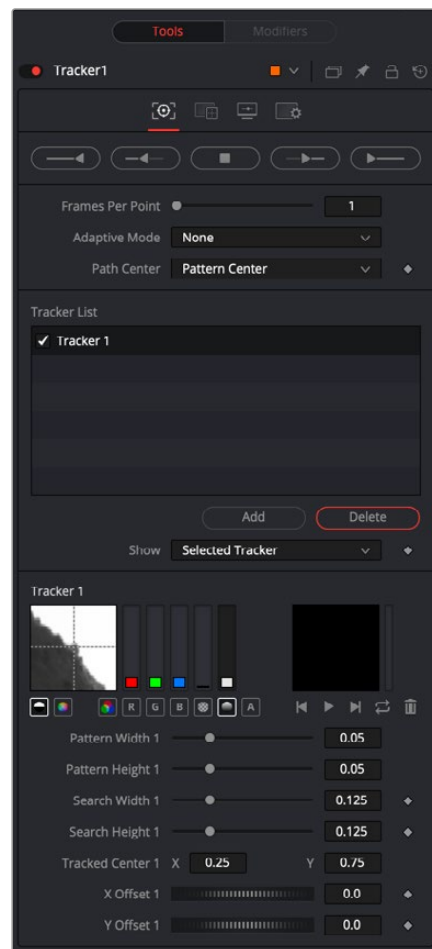


Dragging the handle magnifies the pattern rectangle for precise placement.

TIP: There is no limit to the number of trackers that can be used in one composition, or the number of objects that use the tracking data. There is also no limit to the number of patterns that can be tracked by a single Tracker node. This chapter serves as a reference for the various controls in the Tracker, but we strongly suggest you read the more general information see *Chapter 82, "Using the Tracker Node,"* in the DaVinci Resolve Reference Manual or Chapter 22 in the Fusion Reference Manual.

The Tracker can be employed in two forms: as a node in the Node Editor or as a modifier attached to a parameter. When used as a node in the Node Editor, the image tracked comes from the input to the Tracker node. When used as a modifier, controls appear in the Modifiers tab for the node with the connected control. Tracker Modifiers can track only one pattern, but the image source can come from anywhere in the composition. Use this technique when tracking a quick position for an element.

Inspector



The Tracker node Trackers tab

Trackers Tab

The Trackers tab contains controls for creating, positioning, and initiating tracking operations. After tracking, offset controls are used to improve the alignment of the image following the track.

Track Buttons

There are four buttons to initiate tracking, and one button in the middle used to stop a track in progress. These buttons can track the current pattern forward or backward in time. Holding the pointer over each button displays a tooltip with the name of each button.

The buttons operate as follows:

- **Track Reverse:** Clicking this button causes all active trackers to begin tracking, starting at the end of the render range and moving backward through time until the beginning of the render range.
- **Track Reverse From Current Time:** Clicking this button causes all active trackers to begin tracking, starting at the current frame and moving backward through time until the beginning of the render range.
- **Stop Tracking:** Clicking this button or pressing ESC stops the tracking process immediately. This button is active only when tracking is in process.

- **Track Forward From Current Time:** Clicking this button causes all active trackers to begin tracking, starting at the current frame and moving forward through time until the end of the render range.
- **Track Forward:** Clicking this button causes all active trackers to begin tracking, starting at the first frame in the render range and moving forward through time until the end of the render range.

Tracking Behavior Controls

The following controls all affect how trackers adapt to changing patterns, how the resulting track path is defined, and how many keyframes are generated.

Frames Per Path Point

This slider determines how often the Tracker sets a keyframe on the path. The default is 1, which sets a keyframe on the tracked path at every frame.

Increasing the value causes the tracked path to be less accurate. This may be desirable if the track is returning fluctuating results, but under normal circumstances, leave this control at its default value.

TIP: If the project is field rendered, a value of 1 sets a keyframe on every field. Since the Tracker is extremely accurate, this will result in a slight up-and-down jittering due to the position of the fields. For better results when tracking interlaced footage in Field mode, set the Frames Per Path Point slider to a value of 2, which results in one keyframe per frame of your footage.

Adaptive Mode

Fusion is capable of reacquiring the tracked pattern, as needed, to help with complex tracks. This menu determines the Adaptive tracking method.

- **None:** When set to None, the tracker searches for the original pattern in each frame.
- **Every Frame:** When set to Every Frame, the tracker reacquires the pattern every frame. This helps the Tracker compensate for gradual changes in profile and lighting over time.
- **Best Match:** When set to Best Match, the tracker compares the original selected pattern to the pattern acquired at each frame. If the variation between the two patterns exceeds the threshold amount defined by the Match Tolerance control, the tracker does not reacquire the pattern on that frame. This helps to avoid Tracker drift caused by transient artifacts that cross the pattern's path (such as a shadow).

Path Center

This menu determines how the Tracker behaves when repositioning a pattern. This menu is particularly useful when a pattern leaves the frame or changes so significantly that it can no longer be tracked.

- **Pattern Center:** When Pattern Center is selected in the menu, the tracked path continues from the center of the new path. This is appropriate when replacing an existing path entirely.
- **Track Center (append):** When Track Center (append) is selected in the menu, the path tracked by a new pattern will be appended to the existing path. The path created is automatically offset by the required amount. This setting is used to set a new tracking pattern when the original pattern moves out of the frame or gets obscured by other objects. This technique work bests if the new pattern is located close to the position of the original pattern to avoid any problems with parallax or lens distortion.

Tracker List

A Tracker node can track multiple patterns. Each tracker pattern created in the current Tracker node is managed in the Tracker List.

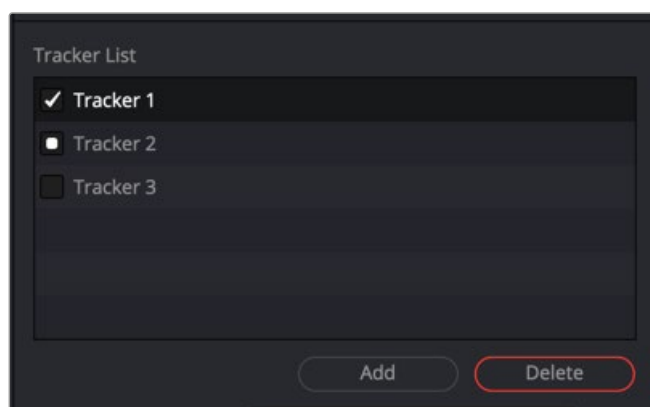
Tracker List

The Tracker List shows the names of all trackers created.

- Each tracker pattern appears in the list by name, next to a small checkbox. Clicking the name of the tracker pattern will select that tracker pattern.
- The controls below the list will change to affect that tracker pattern only. Click a selected tracker pattern once to rename the tracker pattern to something more descriptive.
- Clicking the checkbox changes the state of the tracker.

Tracker States

- **Enabled (black checkbox):** An enabled pattern will re-track each time the track is initiated. Its path data is available for use by other nodes, and the data is available for Stabilization and Corner Positioning.
- **Suspended (white circle):** A Suspended pattern does not re-track when the track is initiated. The data is locked to prevent additional changes. The data from the path is still available for other nodes, and the data is available for advanced Tracking modes like Stabilization and Corner Positioning.
- **Disabled (clear):** A Disabled pattern does not create a path when tracking is initialized, and its data is not available to other nodes or for advanced Tracking operations like Stabilization and Corner Positioning.



Three tracking patterns from top to bottom: enabled, suspended, and disabled

Add/Delete Tracker

Use these buttons to add or delete IntelliTrack or Point trackers from your Tracker List.

Show

This menu selects what controls are displayed in the Tracker node controls. They do not affect the operation of the tracker; they only affect the lower half of the Inspector interface.

- **Selected Tracker Details:** When Selected Tracker Details is chosen, the controls displayed pertain only to the currently selected tracker. You will have access to the Pattern window and the Offset sliders.
- **All Trackers:** When All Trackers is selected, the pattern window for each of the added tracking patterns is displayed simultaneously below the Tracker List.

Left Pattern Display

The pattern display has two side-by-side image windows and a series of status bars. The window on the left shows the pattern initially selected, while the one on the right shows a real-time display of the current pattern as tracking progresses.

As the onscreen controls move while tracking, the display in the leftmost window updates to show the pattern. As the pattern moves, the vertical bars immediately to the right of the image indicate the clarity and contrast of the image channels.

The best channel or channels get selected for tracking based on clarity. These channels have a gray background in the vertical bar representing that channel. You can use the automatic tracking or override the selection and choose the channel by selecting the button beneath the channel to track.



Tracker pattern display

Under normal circumstances, the channel selected shows in the pattern display. If the selected channel is blue, then a grayscale representation of the blue channel for the pattern appears. The image is represented in color only when you activate the Full Color button.

Override this behavior by selecting the Show Full Color button beneath the pattern display instead of the Show Selected Channel button.

TIP: Because Fusion looks for the channel with the highest contrast automatically, you might end up tracking a noisy but high-contrast channel. Before tracking, it's always a good idea to zoom in to your footage and check the RGB channels individually.

Right Pattern Display

The pattern display on the right indicates the actual pattern acquired for tracking. This display is black until tracking the selected pattern for the first time. The pattern display becomes active during tracking, displaying the pattern that Fusion acquires from frame to frame.

As the tracking occurs, the pattern from each frame accumulates into a Flipbook, which can be played back in the pattern window after tracking by using the transport controls at the bottom of the window.

While the track is progressing, the vertical bar immediately to the right of the pattern shows how confident Fusion is that the current pattern matches the initially selected pattern. A green bar indicates a high degree of confidence that the current pattern matches the original, a yellow bar indicates less certainty, and a red bar indicates extreme uncertainty.

After tracking, the pattern display shows a small Flipbook of the track for that pattern to help identify problem frames for the track.

Tracker Sizes

In addition to onscreen controls, each tracker has a set of sizing parameters that let you adjust the pattern and search box.

- **Pattern Width and Height:** Use these controls to adjust the width and height of the selected tracker pattern manually. The size of the tracker pattern can also be adjusted in the viewer, which is the normal method, but small adjustments are often easier to accomplish with the precision of manual controls.
- **Search Width and Height:** The search area defines how far Fusion will look in the image from frame to frame to reacquire the pattern during tracking. As with the Pattern Width and Height, the search area can be adjusted in the viewer, but you may want to make small adjustments manually using these controls.

Tracked Center

This positional control indicates the position of the tracker's center. To remove a previously tracked path from a tracker pattern, right-click this parameter and select Remove Path from the contextual menu.

X and Y Offset

The Offset controls help to create a track for objects that may not provide very well defined or reliable patterns. The Offset controls permit the tracking of something close to the intended object instead. Use these Offsets to adjust the desired position of the path, while the tracker pattern rectangle is positioned over the actual tracking location.

The Offset can also be adjusted directly in the viewer by activating the Offsets button in the viewer toolbar.

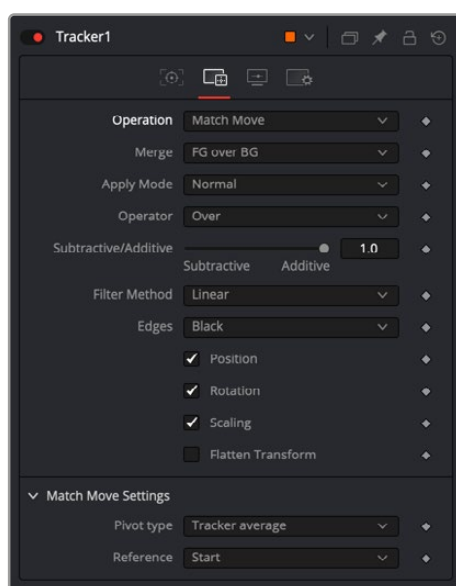


The tracker offset icon in the upper left of the viewer is used to offset the tracking pattern from the intended object.

Operation Tab

While the Trackers tab controls let you customize how the Tracker node analyzes motion to create motion paths, the Operation tab puts the analyzed motion data to use, performing image transforms of various kinds.

The Tracker node is capable of performing a wide variety of functions, from match moving an object into a moving scene, smoothing out a shaky camera movement, or replacing the content of a sign. Use the options and buttons in the Operation tab to select the function performed by the Tracker node.



The Tracker Operation tab

Operation Menu

The Operation menu contains four functions performed by the Tracker. The remaining controls in this tab fine-tune the result of this selection.

- **None:** The Tracker performs no additional operation on the image beyond merely locating and tracking the chosen pattern. This is the default mode, used to create a path that will then drive another parameter on another node.
- **Match Move:** When only the orange background input is connected, this mode stabilizes the image. When a foreground image is connected to the green foreground input, the foreground image matches the position, rotation, and scaling based on the tracking patterns. Stabilizing and match move require a minimum of one tracking pattern to determine position, and two or more to determine scaling and rotation.
- **Corner Positioning:** The Corner Positioner mode tracks the four corners of a rectangular object and replaces the contents with a new image. This function requires a minimum of four tracking patterns. If there are not enough tracking patterns, new tracking patterns are added until the total equals four.
- **Perspective Positioning:** This mode is the inverse of the Corner Positioning mode. Rather than replacing the contents of the rectangle, the four trackers are mapped to the four corners of the image. This is generally used to remove perspective from an image. Like the Corner Positioning mode, this mode requires four tracking patterns, which automatically get added if there are fewer patterns.

Additional Layering Controls

When you choose any operation other than None, a series of additional controls appear.

Merge

The Merge control determines what is done (if anything) with the image provided to the green Foreground input of the Tracker. This menu appears when the operation is set to anything other than None.

- **BG Only:** The foreground input is ignored; only the background is affected. This is used primarily when stabilizing the background image.
- **FG Only:** The foreground input is transformed to match the movement in the background, and this transformed image is passed through the Tracker's output. This Merge technique is used when match moving one layer's motion to another layer's motion.
- **FG Over BG:** The foreground image is merged over the background image, using the Merge method described by the Apply Mode control that appears.
- **BG Over FG:** The background is merged over the foreground. This technique is often used when tracking a layer with an Alpha channel so that a more static background can be applied behind it.

Apply Mode and Operator Menus

This menu provides a variety of options that determine how the two layers should be combined. The options in this menu are identical to those found in the Merge node.

- **Apply Modes:** The Apply Mode setting determines the math used when blending or combining the foreground and background pixels.

Normal: The Default merge mode uses the foreground's alpha channel as a mask to determine which pixels are transparent and which are not. When this is active, another menu shows possible operations, including Over, In, Held Out, Atop, and XOr.

Screen: Screen merges the images based on a multiplication of their color values. The alpha channel is ignored, and layer order becomes irrelevant. The resulting color is always lighter. Screening with black leaves the color unchanged, whereas screening with white will always produce white. This effect creates a similar look to projecting several film frames onto the same surface. When this is active, another menu shows possible operations, including Over, In, Held Out, Atop, and XOr.

Dissolve: Dissolve mixes two image sequences together. It uses a calculated average of the two images to perform the mixture.

Multiply: Multiplies the values of a color channel. This will give the appearance of darkening the image as the values are scaled from 0 to 1. White has a value of 1, so the result would be the same. Gray has a value of 0.5, so the result would be a darker image or, in other words, an image half as bright.

Overlay: Overlay multiplies or screens the color values of the foreground image, depending on the color values of the background image. Patterns or colors overlay the existing pixels while preserving the highlights and shadows of the color values of the background image. The background image is not replaced but is mixed with the foreground image to reflect the original lightness or darkness of the background image.

Soft Light: Soft Light darkens or lightens the foreground image, depending on the color values of the background image. The effect is similar to shining a diffused spotlight on the image.

Hard Light: Hard Light multiplies or screens the color values of the foreground image, depending on the color values of the background image. The effect is similar to shining a harsh spotlight on the image.

Color Dodge: Color Dodge uses the foreground's color values to brighten the background image. This is similar to the photographic practice of dodging by reducing the exposure of an area of a print.

Color Burn: Color Burn uses the foreground's color values to darken the background image. This is similar to the photographic practice of burning by increasing the exposure of an area of a print.

Darken: Darken looks at the color information in each channel and selects the background or foreground image's color value, whichever is darker, as the result color. Pixels lighter than the merged colors are replaced, and pixels darker than the merged color do not change.

Lighten: Lighten looks at the color information in each channel and selects the background or foreground image's color values, whichever is lighter, as the result color value. Pixels darker than the merged color are replaced, and pixels lighter than the merged color do not change.

Difference: Difference looks at the color information in each channel and subtracts the foreground color values from the background color values or the background from the foreground, depending on which has the greater brightness value. Merging with white inverts the color. Merging with black produces no change.

Exclusion: Exclusion creates an effect similar to but lower in contrast than the Difference mode. Merging with white inverts the base color values. Merging with black produces no change.

Hue: Hue creates a result color with the luminance and saturation of the background color values and the hue of the foreground color values.

Saturation: Saturation creates a result color with the luminance and hue of the base color and the saturation of the blend color.

Color: Color creates a result color with the luminance of the background color value and the hue and saturation of the foreground. This preserves the gray levels in the image and is useful for coloring monochrome images.

Luminosity: Luminosity creates a color with the hue and saturation of the background color and the luminance of the foreground color. This mode creates an inverse effect from that of the Color mode.

- **Operator Modes:** This menu is used to select the Operation Mode of the merge. It determines how the foreground and background are combined to produce a result. This drop-down menu is visible only when the Merge node's Apply Mode is set to either Normal or Screen.

NOTE: For an excellent description of the math underlying the Operation modes, read "Compositing Digital Images," Porter, T., and T. Duff, SIGGRAPH 84 proceedings, pages 253-259. Essentially, the math is as described below.

TIP: Some modes not listed in the Operator drop-down menu (Under, In, Held In, Below) are easily obtained by swapping the foreground and background inputs and choosing a corresponding mode.

The formula used to combine pixels in the merge is always $fg * x + bg * y$. The different operations determine exactly what x and y are, as shown in the description for each mode.

The Operator modes are as follows:

- **Over:** The Over mode adds the foreground layer to the background layer by replacing the pixels in the background with the pixels from the Z wherever the foreground's alpha channel is greater than 1.

$$x = 1, y = 1 - [\text{foreground Alpha}]$$

- **In:** The In mode multiplies the alpha channel of the background input against the pixels in the foreground. The color channels of the foreground input are ignored. Only pixels from the foreground are seen in the final output. This essentially clips the foreground using the mask from the background.

$$x = [\text{background Alpha}], y = 0$$

- **Held Out:** Held Out is essentially the opposite of the In operation. The pixels in the foreground image are multiplied against the inverted alpha channel of the background image. You can accomplish exactly the same result using the In operation and a Matte Control node to invert the matte channel of the background image.

$$x = 1 - [\text{background Alpha}], y = 0$$

- **ATop:** ATop places the foreground over the background only where the background has a matte.

$$x = [\text{background Alpha}], y = 1 - [\text{foreground Alpha}]$$

- **XOr:** XOr combines the foreground with the background wherever either the foreground or the background have a matte, but never where both have a matte.

$$x = 1 - [\text{background Alpha}], y = 1 - [\text{foreground Alpha}]$$

- **Subtractive/Additive:** This slider controls whether Fusion performs an Additive merge, a Subtractive merge, or a blend of both. This slider defaults to Additive merging for most operations, assuming the input images are premultiplied (which is usually the case). If you don't understand the difference between Additive and Subtractive merging, here's a quick explanation:

- An Additive merge is necessary when the foreground image is premultiplied, meaning that the pixels in the color channels have been multiplied by the pixels in the alpha channel. The result is that transparent pixels are always black since any number multiplied by 0 always equals 0. This obscures the background (by multiplying with the inverse of the foreground alpha), and then simply adds the pixels from the foreground.
- A Subtractive merge is necessary if the foreground image is not premultiplied. The compositing method is similar to an Additive merge, but the foreground image is first multiplied by its alpha to eliminate any background pixels outside the alpha area.

In most software applications, you will find the Additive/Subtractive option displayed as a simple checkbox. Fusion lets you blend between the Additive and Subtractive versions of the merge operation, which is occasionally useful for dealing with problem composites with edges that are calling attention to themselves as too bright or too dark.

For example, using a Subtractive setting on a premultiplied image may result in darker edges. Using an Additive setting with a non-premultiplied image may result in lightening the edges. By blending between Additive and Subtractive, you can tweak the edge brightness to be just right for your situation.

Filter Method (Match Move)

Determines which filter to use to handle image transforms made using the Tracker node. This menu appears only when the Operation Mode is set to Match Move.

- **Box:** This is a simple interpolation resize of the image.
- **Linear:** This uses a simplistic filter, which produces relatively clean and fast results.
- **Quadratic:** This filter produces a nominal result. It offers a good compromise between speed and quality.
- **Cubic:** This produces better results with continuous-tone images but is slower than Bi-Cubic. If the images have fine detail in them, the results may be blurrier than desired.
- **Catmull-Rom:** This produces good results with continuous-tone images that are resized down. It produces sharp results with finely detailed images.
- **Gaussian:** This is very similar in speed and quality to Bi-Cubic.

- **Mitchell:** This is similar to Catmull-Rom but produces better results with finely detailed images. It is slower than Catmull-Rom.
- **Lanczos:** This is very similar to Mitchell and Catmull-Rom but is a little cleaner and also slower.
- **Sinc:** This is an advanced filter that produces very sharp, detailed results; however, it may produce visible “ringing” in some situations.
- **Bessel:** This is similar to the Sinc filter but may be slightly faster.

Edges

This menu selects how the revealed edges are handled when the image is moved to match position and scaling.

- **Black Edges:** Out-of-frame edges revealed by Stabilization are left black.
- **Wrap:** Portions of the image moved off frame to one side are used to fill edges that are revealed on the opposite side.
- **Duplicate:** The last valid pixel on an edge is repeated to the edge of the frame.
- **Mirror:** Image pixels are mirrored to fill to the edge of the frame.

Position, Rotation, and Scaling Checkboxes (Match Move)

The Position, Rotation, and Scaling checkboxes appear only when the mode is set to Match Move. They determine what components of motion that Stabilization will attempt to correct in the image. For example, if only the Position checkbox is selected, no attempt will be made to correct for Rotation and Scaling in the image.

Flatten Transformation (Match Move)

This checkbox appears only when the mode is set to Match Move. Like most transformations in Fusion, Stabilization is concatenated with other sequential transformations by default. Selecting this checkbox will flatten the transform, breaking any concatenation taking place and applying the transform immediately.

Mapping Type

The Mapping Type control appears only in the Corner Positioning mode. There are two options in the menu:

- **Bi_Linear:** The first method is Bi-Linear, where the foreground image is mapped into the background without any attempt to correct for perspective distortion. This is identical to how previous versions of Fusion operated.
- **Perspective:** The foreground image is mapped into the background taking perspective distortion into account. This is the preferred setting since it maps better to the real world than the older Bi-Linear setting.

Corner Selector (Corner or Perspective Positioning)

When the operation of the Tracker is set to either Corner or Perspective Positioning modes, four drop-down menus appear. These options choose which trackers map to each of the four corners of the rectangle. This is useful when a Tracker has more than four patterns selected, and you must choose which patterns the positioners use.

Rotate Clockwise and Counter-Clockwise Buttons (Corner or Perspective Positioning)

These controls appear only when the operation of the Tracker is set to either Corner or Perspective Positioning modes. They are used to rotate the foreground image by 90 degrees before it is applied to the background.

Stabilize Settings

The Tracker node automatically outputs several steady and unsteady position outputs to which other controls in the Node Editor can be connected. The Stable Position output provides X and Y coordinates to match or reverse motion in a sequence. These controls are available even when the operation is not set to Match Move, since the Stable Position output is always available for connection to other nodes.

Match Move Settings

These settings determine how tracking data is correlated with the reference pattern for making transforms.

Pivot Type

The Pivot type menu determines how the anchor point for rotation is selected.

- **Tracker Average:** Averages the location based on the tracking points.
- **Selected Tracker:** Provides a menu where one of the current trackers can be selected as the pivot point.
- **Manual:** Displays X and Y position number fields where you can manually position the pivot points.

Reference

The Reference mode determines the “snapshot frame” based on the frame where the pattern is first selected. All Stabilization is intended to return the image back to that reference.

- **Select Time:** Lets you select the current frame.
- **Start:** The Snapshot Frame is determined to be the first frame in the tracked path. All Stabilization is intended to return the image back to that reference.
- **Start and End:** The Start and End Reference mode is somewhat different from all other Reference modes. Where the others are intended to take a snapshot frame to which all stabilization returns, immobilizing the image, the Start and End mode is intended to smooth existing motion, without removing it. This mode averages the motion between the Start and End of the path, drawing a straight line between those points.

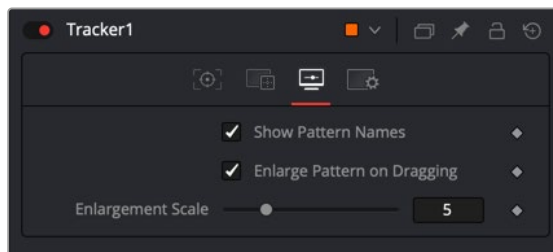
When this mode is active, it reveals the Reference Intermediate Points control. Increasing the value of this control increases the number of points in the path used by the Reference, smoothing the motion from a straight line between Start and End without making it wholly linear.

- **End:** The Snapshot Frame is determined to be the last frame in the tracked path. All Stabilization is intended to return the image back to that reference.

TIP: By default, the Tracker displays a single displacement path of the tracked data in the Spline Editor. To view X and Y paths of the tracked points in the Spline Editor, go to Preferences > Globals > Splines.

Display Options Tab

The Display Options tab lets you customize the look of onscreen controls in the viewer.



The Tracker Display Options tab

Show Pattern Names

This option defines whether the Tracker's pattern names will be displayed in the viewer. Switch it off to see the pattern rectangle instead.

Enlarged Pattern on Dragging

This option defines whether there is a magnified thumbnail view when positioning the pattern rectangle.

Enlargement Scale

The zoom factor that is used when positioning the pattern rectangle when the above option is activated. TIP: The outputs of a tracker (seen in the Connect to... menu) can also be used by scripts. They are:

- **SteadyPosition:** Steady Position
- **UnsteadyPosition:** Unsteady Position
- **SteadyAxis:** Steady Axis
- **SteadySize:** Steady Size
- **UnsteadySize:** Unsteady Size
- **SteadyAngle:** Steady Angle
- **UnsteadyAngle:** Unsteady Angle
- **Position1:** Tracker 1 Offset position
- **PerspectivePosition1:** Tracker 1 Perspective Offset position
- **PositionX1:** Tracker 1 Offset X position (3D Space)
- **PositionY1:** Tracker 1 Offset Y position (3D Space)
- **PerspectivePositionX1:** Tracker 1 Perspective Offset X position (3D Space)
- **PerspectivePositionY1:** Tracker 1 Perspective Offset Y position (3D Space)
- **SteadyPosition1:** Tracker 1 Steady Position
- **UnsteadyPosition1:** Tracker 1 Unsteady Position (likewise for the 2nd, 3rd, and so on)

Common Controls

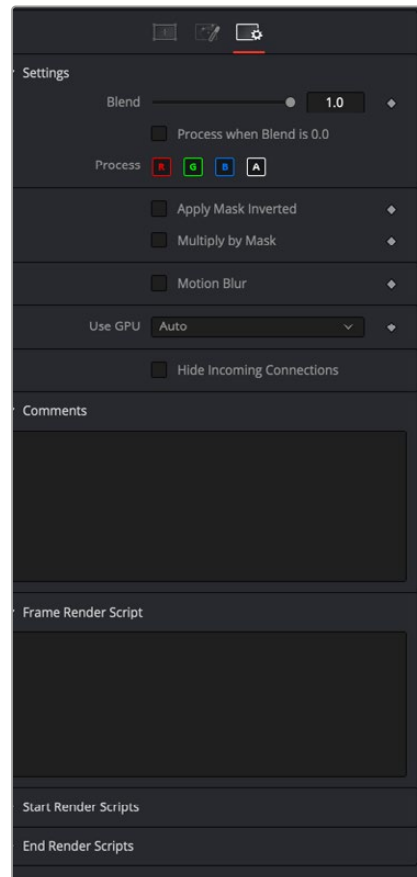
Settings Tab

The Settings tab in the Inspector is also duplicated in other Tracking nodes. These common controls are described in detail at the end of this chapter in "The Common Controls" section.

The Common Controls

Nodes that handle tracking operations share several identical controls in the Inspector. This section describes controls that are common among Tracking nodes.

Inspector



The Common Tracking Controls Settings tab

Settings Tab

The Settings tab in the Inspector can be found on every tool in the Tracking category. The controls are consistent and work the same way for each tool.

Blend

The Blend control is used to blend between the tool's original image input and the tool's final modified output image. When the blend value is 0.0, the outgoing image is identical to the incoming image. Normally, this will cause the tool to skip processing entirely, copying the input straight to the output.

Process When Blend Is 0.0

The tool is processed even when the input value is zero. This can be useful if the node is scripted to trigger a task, but the node's value is set to 0.0.

Red/Green/Blue/Alpha Channel Selector

These four buttons are used to limit the effect of the tool to specified color channels. This filter is often applied after the tool has been processed.

For example, if the Red button on a Blur tool is deselected, the blur will first be applied to the image, and then the red channel from the original input will be copied back over the red channel of the result.

There are some exceptions, such as tools for which deselecting these channels causes the tool to skip processing that channel entirely. Tools that do this will generally possess a set of identical RGBA buttons on the Controls tab in the tool. In this case, the buttons in the Settings and the Controls tabs are identical.

Apply Mask Inverted

Enabling the Apply Mask Inverted option inverts the complete mask channel for the tool. The mask channel is the combined result of all masks connected to or generated in a node.

Multiply by Mask

Selecting this option will cause the RGB values of the masked image to be multiplied by the mask channel's values. This will cause all pixels of the image not included in the mask (i.e., set to 0) to become black/transparent.

Use Object/Use Material (Checkboxes)

Some 3D software can render to file formats that support additional channels. Notably, the EXR file format supports Object ID and Material ID channels, which can be used as a mask for the effect. These checkboxes determine whether the channels will be used, if present. The specific Material ID or Object ID affected is chosen using the next set of controls.

Correct Edges

This checkbox appears only when the Use Object or Use Material checkboxes are selected. It toggles the method used to deal with overlapping edges of objects in a multi-object image. When enabled, the Coverage and Background Color channels are used to separate and improve the effect around the edge of the object. If this option is disabled (or no Coverage or Background Color channels are available), aliasing may occur on the edge of the mask.

For more information on the Coverage and Background Color channels, see *Chapter 78, "Understanding Image Channels,"* in the DaVinci Resolve Reference Manual, or Chapter 18 in the Fusion Reference Manual.

Object ID/Material ID (Sliders)

Use these sliders to select which ID will be used to create a mask from the object or material channels of an image. Use the Sample button in the same way as the Color Picker: to grab IDs from the image displayed in the viewer. The image or sequence must have been rendered from a 3D software package with those channels included.

Motion Blur

- **Motion Blur:** This toggles the rendering of Motion Blur on the tool. When this control is toggled on, the tool's predicted motion is used to produce the motion blur caused by the virtual camera's shutter. When the control is toggled off, no motion blur is created.
- **Quality:** Quality determines the number of samples used to create the blur. A quality setting of 2 will cause Fusion to create two samples to either side of an object's actual motion. Larger values produce smoother results but increase the render time.

- **Shutter Angle:** Shutter Angle controls the angle of the virtual shutter used to produce the motion blur effect. Larger angles create more blur but increase the render times. A value of 360 is the equivalent of having the shutter open for one full frame exposure. Higher values are possible and can be used to create interesting effects.
- **Center Bias:** Center Bias modifies the position of the center of the motion blur. This allows for the creation of motion trail effects.
- **Sample Spread:** Adjusting this control modifies the weighting given to each sample. This affects the brightness of the samples.

Use GPU

The Use GPU menu has three settings. Setting the menu to Disable turns off GPU hardware-accelerated rendering. Enabled uses the GPU hardware for rendering the node. Auto uses a capable GPU if one is available and falls back to software rendering when a capable GPU is not available.

Hide Incoming Connections

Enabling this checkbox can hide connection lines from incoming nodes, making a node tree appear cleaner and easier to read. When enabled, empty fields for each input on a node will be displayed in the Inspector. Dragging a connected node from the node tree into the field will hide that incoming connection line as long as the node is not selected in the node tree. When the node is selected in the node tree, the line will reappear.

Comments

The Comments field is used to add notes to a tool. Click in the empty field and type the text. When a note is added to a tool, a small red square appears in the lower-left corner of the node when the full tile is displayed, or a small text bubble icon appears on the right when nodes are collapsed. To see the note in the Node Editor, hold the mouse pointer over the node to display the tooltip.

Scripts

Three Scripting fields are available on every tool in Fusion from the Settings tab. They each contain edit boxes used to add scripts that process when the tool is rendering. For more details on scripting nodes, please consult the Fusion scripting documentation.